ETAS RTA-BSW v12.7.0

RTA-Fray Stack Reference Guide

www.etas.com

Copyright

The data in this document may not be altered or amended without special notification from ETAS GmbH. ETAS GmbH undertakes no further obligation in relation to this document. The software described in it can only be used if the customer is in possession of a general license agreement or single license. Using and copying is only allowed in concurrence with the specifications stipulated in the contract.

Under no circumstances may any part of this document be copied, reproduced, transmitted, stored in a retrieval system or translated into another language without the express written permission of ETAS GmbH.

© Copyright 2025 ETAS GmbH, Stuttgart

The names and designations used in this document are trademarks or brands belonging to the respective owners.

ETAS RTA-BSW 12.7.0 - RTA-Fray Stack Reference Guide R01 EN – 05.2025

Contents

1 FlexRay Interface . . . 5
- 1.1 Introduction . . . 5
- 1.2 Supported Features . . . 6
- 1.3 Extensions . . . 8
- 1.4 Limitations . . . 19
- 1.5 Exclusions . . . 19
- 1.6 Deviations . . . 20
- 1.7 API Definition . . . 25
- 1.8 Configuration Advice . . . 27
- 1.9 Integration Advice . . . 28
- 1.10 Hardware Requirements . . . 29

2 FlexRay State Manager . . . 30
- 2.1 Introduction . . . 30
- 2.2 Supported Features . . . 31
- 2.3 Extensions . . . 32
- 2.4 Limitations . . . 34
- 2.5 Exclusions . . . 34
- 2.6 Deviations . . . 34
- 2.7 API Definition . . . 35
- 2.8 Configuration Advice . . . 35
- 2.9 Integration Advice . . . 37
- 2.10 Hardware Requirements . . . 38

3 FlexRay Network Management . . . 39
- 3.1 Introduction . . . 39
- 3.2 Supported Features . . . 40
- 3.3 Extensions . . . 41
- 3.4 Limitations . . . 46
- 3.5 Exclusions . . . 46
- 3.6 Deviations . . . 47
- 3.7 API Definition . . . 48
- 3.8 Configuration Advice . . . 49
- 3.9 Integration Advice . . . 53
- 3.10 Hardware Requirements . . . 55

4 FlexRay Transport Protocol 56
4.1 Introduction 56
4.2 Supported Features 57
4.3 Extensions 57
4.4 Limitations 61
4.5 Exclusions 61
4.6 Deviations 61
4.7 API Definition 61
4.8 Configuration Advice 62
4.9 Integration Advice 68
4.10 Hardware Requirements 69
5 Glossary 70

1 FlexRay Interface

Module name	FrIf
Package	RTA-FRAY
AUTOSAR Module ID	61
AUTOSAR Version	22-11

1.1 Introduction

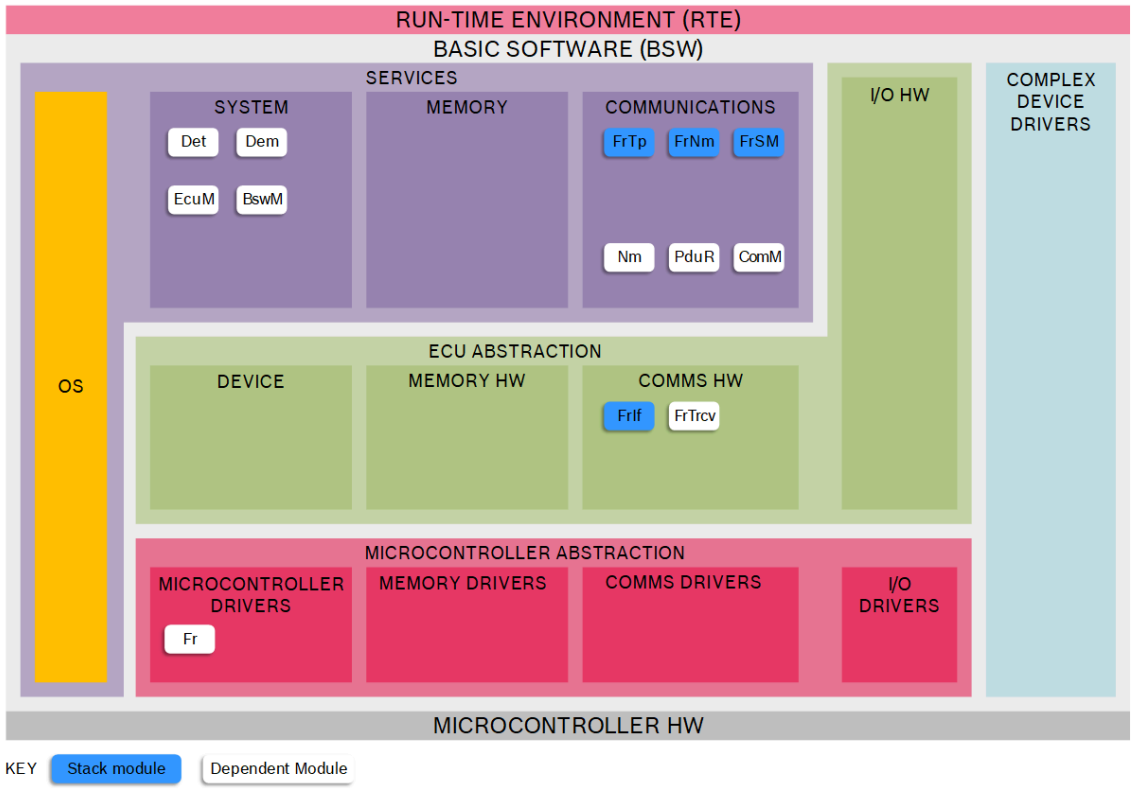


Figure 1.1.FlexRay Architecture

The FlexRay Interface module (FrIf) is located in the communication hardware abstraction layer, between the communication drivers layer and the upper communication services layer. The diagram above shows the position of FrIf in the AUTOSAR layered architecture.

FrIf accesses multiple communication controllers in the microcontrollers layer via the communication drivers, performing an abstraction of the hardware-specific information and then calling the required driver with the appropriate parameters. Note that FrIf may utilise multiple drivers, each of which may in turn access multiple controllers.

The purpose of FrIf is:

1. To provide a generally-specified interface to the communication system for upper layer communication modules.

2. To offer high data transmission rates and increased fault-tolerance (compared to other communication protocols).
3. To provide both time-triggered and event-triggered communication.
4. To provide synchronous and asynchronous transfer of send (Tx) and receive (Rx) frames.
5. To provide initialisation and clock synchronisation functions for individual device nodes.

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

1.2 Supported Features

1.2.1 Data handling

FrIf supports the following data handling features:

- Frame-based transmit and receive functions (up to 254 bytes of payload data per frame).
- PDU packing (multiple PDUs per frame).
- PDU Update bits (enabling recognition of updated/unchanged PDUs in a frame).
- Decoupled and immediate transmission modes.
- Configurable dynamic PDU payload lengths.

1.2.2 Transmission

FrIf provides a mechanism for PDU transmission via the FrIf_Transmit function.

- Checks the initialisation status of FrIf, and only executes if Fr_Init has already been called.
- Executes decoupled transmissions from the upper software layer asynchronously (independent of FlexRay global time) through the communication cycle job list.
- May be invoked directly from upper layer modules for immediate (non-buffered) transmission of a single PDU.
- Makes a call (using Fr_TransmitTxLPdu) to the FlexRay driver to perform the transmission operation. If E_OK is returned, the transmission is successfully completed.
- If transmission confirmation is configured (through FrIfConfirm), a confirmation will be requested through FrIf_ProvideTxConfirmation.
- Continuous transmission mode (required by some FlexRay CCs) is supported via the FrIfAlwaysTransmit configuration parameter.

1.2.3 Reception

FrIf supports the following interface functions for processing received PDUs:

- FrIf_RxAndIndicate processes all frames in the current job list and checks for PDUs with an update bit set.
- If true, or if no update bit has been configured, the upper layer function (UL_RxIndication) is called with the PDU Id and a reference to the structure containing the length of the received PDU and a pointer to the PDU data. If the update bit is not set, the PDU is ignored.

1.2.4 Communication Operations

FrIf currently supports the following Communication Operations:

- ReceiveAndIndicate
- TransmitWithDecoupledBufferAccess
- ProvideTxConfirmation

1.2.5 Padding

Three kinds of padding are currently supported.

1.2.5.1 Padding Bytes in PDUs (Applicable for trigger transmit Pdus)

This involves the padding of PDUs which are partially filled by the upper layers during a trigger transmit call. This padding is applicable at the PDU level. For example, FrIf_Transmit requests a trigger transmit PDU with a configured length as 16. The trigger transmit call returns a length of 12, and the remaining 4 bytes are padded with a default value (if configured).

1.2.5.2 Zero payload transmission (Applicable for multiple PDUs with update bits)

This involves the transmission of a zero data for PDUs which are not requested for transmission, but the frame has to be transmitted (i.e. other PDUs in the frame have been requested for transmission). For example, a frame has 2 PDUs with update bits configured with a Frame length of 16. PDU 1 of length 8 is requested for transmission, the frame is transmitted with PDU 2 filled with a zero value (if configured).

1.2.5.3 FrIfUnusedBitValue (Applicable for multiple PDUs)

This involves the transmission of a default configured value for PDUs which are not requested for transmission but where the frame has to be transmitted (i.e other PDUs in the frame have been requested for transmission). Along with this, the unused bits in the updatebit byte have to be filled with the default configured value. For example, a frame has 2 PDUs with update bits configured with a frame length of 16. PDU 1 of length 8 is requested for transmission, the frame is transmitted with PDU 2 filled with the default configured bit value of 1. Update bits 127

for PDU 1 and 126 for PDU 2 configured. Hence the updatebit byte value = 0x3F.

1.2.6 Dual Controller Configuration

FrIf supports dual controller configuration which can be used to support more than 128 frames. The wakeup or coldstart actions are performed based on the FrController configured via FrIfFrCtrlRef to FrIf controller 0. Therefore, from the FrIf point of view there will only be one controller. The allowed Frame configurations are:

- Controller 1 has only Tx frames and Controller 2 has Tx and Rxframes
- Controller 1 has Tx and Rx frames and Controller 2 has only Rxframes
- Controller 1 has Rx frames and Controller 2 has Rx frames
- Both controllers can have both Tx and Rx frames

A dummy frame will not be generated because the Fr takes the index for each of the FrIfFrameTriggerings from the FrIfLPdu container.

If any one of the controllers is not Normally Active, then FrSM has to initiate the startup procedure.

1.2.7 Scheduling (Job Lists)

FrIf supports scheduling in the form of Job Lists.

Each job list contains a set of communication operations, sorted by execution order. The list is processed by a dedicated execution function (initiated by a FlexRay interrupt) which is associated to one specific Communication Controller (CC). This means that there will be one list and one execution function instance per CC, up to the configured maximum CC count.

Note that transmission operations through the Job List are always performed with decoupled buffer access. All operations in the job list are allocated a specific time slot for execution.

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_ACS_CH_A

Short Name	FrIfRbDynCreate_FRIF_E_ACS_CH_A

Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRIF_E_ACS_CH_A) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.2 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_ACS_CH_B

Short Name	FrIfRbDynCreate_FRIF_E_ACS_CH_B
Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRIF_E_ACS_CH_B) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.3 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_NIT_CH_A

Short Name	FrIfRbDynCreate_FRIF_E_NIT_CH_A
Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRIF_E_NIT_CH_A) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.4 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_NIT_CH_B

Short Name	FrIfRbDynCreate_FRIF_E_NIT_CH_B
Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRIF_E_NIT_CH_B) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.5 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_SW_CH_A

Short Name	FrIfRbDynCreate_FRIF_E_SW_CH_A
Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FrIfRbDynCreate_FRIF_E_SW_CH_A) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.6 FrIf/FrIfConfig/FrIfCluster/FrIfClusterDemEventParameterRefs/ FrIfRbDynCreate_FRIF_E_SW_CH_B

Short Name	FrIfRbDynCreate_FRIF_E_SW_CH_B
Description	Selects the way how the FrIf DemEvents are created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FrIfRbDynCreate_FRIF_E_SW_CH_B) is also automatically created and shall therefore not be set manually.

Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.7 Frlf/FrlfConfig/FrlfCluster/FrlfController/FrlfAbsTimer

Short Name	FrlfAbsTimer
Description	This container contains the configuration of an absolute timer of a FlexRay CC.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.8 Frlf/FrlfConfig/FrlfCluster/FrlfController/FrlfAbsTimer/FrlfFrAbsTimerRef

Short Name	FrlfFrAbsTimerRef
Description	Reference to an absolute timer, which is handled by a specific FlexRay Driver. This reference is unique for the ECU, therefore, an explicit reference to the FlexRay Driver is not necessary. Since this parameter is not used, the lower multiplicity value of this parameter is changed to 0. This is to remove the BCT validation warnings.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.9 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrlfJob/FrlfJobListCallbackName

Short Name	FrlfJobListCallbackName
Description	This parameter defines the name of the user callback function name, which needs to be called after the execution of RxIndication and TxConfirmation but before deCouple Transmission.
Multiplicity	0..1
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.10 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard

Short Name	FrifJobListWizard
Description	Container to configure the number of job entries per FlexRay cycle.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.11 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobDelay

Short Name	FrlfJobDelay
Description	Delay between the current job and the next job in the FlexRay joblist.
Multiplicity	1..1
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.12 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry

Short Name	FrlfJobEntry
Description	Entry for the job interrupt in the FlexRay cycle.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.13 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfExecutionOffset

Short Name	FrlfExecutionOffset
Description	This value is the offset provided for the excution of the job entry. It will be used only for job entries in the dynamic segment. Value is configured in macroticks
Multiplicity	0..1
Value Range	0..16000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.14 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfFrames

Short Name	FrlfFrames

Description	This is the expert joblist configuration. This field should be configured by someone who has knowledge in FlexRay. No validation is provided for the additional frames configured here.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.15 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfJobListCallback

Short Name	FrlfJobListCallback
Description	This parameter defines the name of the user callback function name, which will be called after Receive and Indicate, Tx confirmation and before Decoupled Tx. Supports Multiple callbacks FrlfJobListCallback is not supported for the implementation configuration variant Post Build Loadable.
Multiplicity	0..*
Default Value	-
Type	ECUC-FUNCTION-NAME-DEF

1.3.1.16 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfMacroTick

Short Name	FrlfMacroTick
Description	Macrotick offset in the Cycle [Macrotick].
Multiplicity	1..1
Value Range	0..16000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.17 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntryDelayMargin

Short Name	FrlfJobEntryDelayMargin
Description	Safety margin in macroticks to set the absolute timer for the next joblist execution,If we are in current cycle but macrotick for next joblist is already passed
Multiplicity	0..1
Value Range	0..16000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.18 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobListCFGType

Short Name	FrlfJobListCFGType
Description	Note : The joblist algorithm type AR31 name has been replaced by STANDARD
Multiplicity	1..1
Default Value	ESP
Enumeration Literals	ESP, STANDARD
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.19 Frlf/FrlfConfig/FrlfCluster/FrlfJoblistExecDelayCompensation

Short Name	FrlfJoblistExecDelayCompensation
Description	Macroticks to compensate the delay between Fr_GetGlobalTime and Fr_SetAbsoluteTimer APIs in Frlf_JobListExec_<FrlfCluster.ShortName>(). This value should not exceed the value configured in FrlfMaxIsrDelay
Multiplicity	0..1
Value Range	0..10240000
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.20 Frlf/FrlfConfig/FrlfCluster/GClusterDriftDamping

Short Name	GClusterDriftDamping
Description	The cluster drift damping factor, based on the longest microtick gdMaxMicrotick used in the cluster. Used to compute the local cluster drift damping factor pClusterDriftDamping [Micrroticks].
Multiplicity	1..1
Value Range	0..5
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.21 Frlf/FrlfConfig/FrlfCluster/GOffsetCorrectionStart

Short Name	GOffsetCorrectionStart
Description	Start of the offset correction phase within the NIT, expressed as the number of macroticks from the start of cycle [Macroticks].
Multiplicity	1..1

Value Range	7..15999
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.22 FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfRxPdu/FrIfUserTargetRxPduId

Short Name	FrIfUserTargetRxPduId
Description	Defines the target Id of the CDD for Rx Pdus. The current software does not support this parameter.
Multiplicity	0..0
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.23 FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfTxPdu/FrIfSyncTxPduNotification

Short Name	FrIfSyncTxPduNotification
Description	Name of function to be called before I-PDU content is assembled into L-PDU during FrIf JobListExec. Intension: Allow realtime provision of I-PDU data content, reduce latency and avoid sporadic miss of transmission time slot due to jitter in operating system tasks not synchronized to FlexRay.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.24 FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfTxPdu/FrIfUserTargetTxPduId

Short Name	FrIfUserTargetTxPduId
Description	Defines the target Id of the CDD for Tx Pdus. The current software does not support this parameter.
Multiplicity	0..0
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.25 FrIf/FrIfGeneral/FrIfAdditionalFrameCallPosition

Short Name	FrIfAdditionalFrameCallPosition
Description	This parameter will indicate from which position the frames selected for each job entry should be called. This parameter must be configured if there is any additional frames configured in FrIfJobEntry container.If the selected configuration variant is PostBuild Loadable, then it is mandatory to configure this parameter.
Multiplicity	0..1
Default Value	-
Enumeration Literals	FROM_START, FROM_END
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.26 FrIf/FrIfGeneral/FrIfEnableARBackwardCompatibility

Short Name	FrIfEnableARBackwardCompatibility
Description	This parameter enables the support of FrIf with older versions of AUTOSAR
Introduction	true: Enable the support of older AUTOSAR version false: Disable the support of older AUTOSAR version
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.27 FrIf/FrIfGeneral/FrIfMaxBufferSize

Short Name	FrIfMaxBufferSize
Description	Maximum size of the single buffer that is used in case of postbuild loadable configuration. The value should be configured in bytes. FrIf stores the frame data which is aligned to 4 bytes alignment. Because of this, user has to provide the size as per 4 bytes alignment. For ex : if the frame size is configured as 32, then FrIf will need only 8 bytes of uint32. if the frame size is configured as 2, then FrIf will need only 1 bytes of uint32. If this is considered properly, the RAM wastage can be avoided.
Multiplicity	0..1
Value Range	1..*
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.28 FrIf/FrIfGeneral/FrIfMaxFramesCnt

Short Name	FrIfMaxFramesCnt
Description	Maximum number of Frames. This parameter is needed only in case of post-build loadable implementation using static memory allocation.
Multiplicity	0..1
Value Range	1..*
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.29 FrIf/FrIfGeneral/FrIfTransceiverSharing

Short Name	FrIfTransceiverSharing
Description	True: A single transceiver is shared by two controllers with same channel, False: The transceiver is not shared by two controllers.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.30 FrIf/FrIfGeneral/FrIfVendor11Driver

Short Name	FrIfVendor11Driver
Description	Parameter to suppress the dependency between FrIf and Fr
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.31 FrIf/FrIfGeneral/FrIfZeroPayloadTx

Short Name	FrIfZeroPayloadTx
Description	Switch to Transmit Zero payload when there is no IPdu transmit request
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 FrIf_GetSyncState

Syntax	`FrIf_GetSyncState`
Parameters	`FrIf_CtrlIdx` The index which identifies a specific FlexRay Communication Controller.
	`FrIf_SyncStatePtr` A pointer to the structure which will contain the returned state.
Description	Fetches the current clock synchronisation state for a given CC. Returns E_OK if the request is accepted, E_NOT_OK if not accepted.

1.3.2.2 FrIf_SetExtSync

Syntax	`FrIf_SetExtSync`
Parameters	`FrIf_CtrlIdx` The index which identifies a specific FlexRay Communication Controller.
	`FrIf_Offset` The offset correction value to apply.
	`FrIf_Rate` The rate correction value to apply.
Description	Performs an external clock correction for the specified CC using the supplied offset and rate values.

1.3.2.3 FrIf_DeInit

Syntax	`FrIf_DeInit(void)`
Parameters	`Void`
Description	De-Initializes the global variables used in FrIf. Useful when Variant 2 is used.

1.3.2.4 FrIf_RxAndIndicate

Syntax	`FrIf_RxAndIndicate( const uint16 * const msgBufIdxptr_u16, uint16 curMacrotick_u16)`
Description	Loops for all LPDUs and copies data into FrIf buffer if received.

1.3.2.5 FrIf_DecoupledTx

Syntax	`FrIf_DecoupledTx(const uint16 * const msgBufIdxptr_u16)`
Description	Loops for all LPDUs and copies the data into CC buffer for transmission.

1.3.2.6 FrIf_TxConfirmation

Syntax	`FrIf_TxConfirmation( const uint16 * const msgBufIdxptr_u16)`
Description	Loops for all LPDUs and raises Tx Confirmation for the transmitted frames.

1.4 Limitations

The following features are not supported for version v12.7.0:

- Communication operations `Prepare_LPDU` (hardware buffer re-configuration), `ReceiveAndStore` and `ProvideRxIndication` are not supported.
- Only one Transceiver (Id 0) is supported.
- Only one Absolute Timer (Id 0) is supported.

1.5 Exclusions

The following API functions are not currently implemented:

- `FrIf_AckAbsoluteTimerIRQ`
- `FrIf_GetAbsoluteTimerIRQStatus`

The following optional API functions are not included:

- `FrIf_CancelTransmit`
- `FrIf_GetTransceiverError`
- `FrIf_EnableTransceiverBranch`
- `FrIf_DisableTransceiverBranch`
- `FrIf_EnableBusMirroring`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfFrameTriggeringDemEventParameterRefs
- FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfFrameTriggeringDemEventParameterRefs/FRIF_E_LPDU_SLOTSTATUS
- FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfMessageId
- FrIf/FrIfConfig/FrIfCluster/FrIfDetectNITError
- FrIf/FrIfConfig/FrIfCluster/FrIfGColdStartAttempts
- FrIf/FrIfConfig/FrIfCluster/FrIfGCycleCountMax
- FrIf/FrIfConfig/FrIfCluster/FrIfGListenNoise
- FrIf/FrIfConfig/FrIfCluster/FrIfGMaxWithoutClockCorrectFatal
- FrIf/FrIfConfig/FrIfCluster/FrIfGMaxWithoutClockCorrectPassive
- FrIf/FrIfConfig/FrIfCluster/FrIfGSyncFrameIDCountMax
- FrIf/FrIfConfig/FrIfCluster/FrIfGdActionPointOffset
- FrIf/FrIfConfig/FrIfCluster/FrIfGdBit
- FrIf/FrIfConfig/FrIfCluster/FrIfGdCasRxLowMax

- FrIf/FrIfConfig/FrIfCluster/FrIfGdDynamicSlotIdlePhase
- FrIf/FrIfConfig/FrIfCluster/FrIfGdIgnoreAfterTx
- FrIf/FrIfConfig/FrIfCluster/FrIfGdMiniSlotActionPointOffset
- FrIf/FrIfConfig/FrIfCluster/FrIfGdSampleClockPeriod
- FrIf/FrIfConfig/FrIfCluster/FrIfGdSymbolWindow
- FrIf/FrIfConfig/FrIfCluster/FrIfGdSymbolWindowActionPointOffset
- FrIf/FrIfConfig/FrIfCluster/FrIfGdTSSTransmitter
- FrIf/FrIfConfig/FrIfCluster/FrIfGdWakeupRxIdle
- FrIf/FrIfConfig/FrIfCluster/FrIfGdWakeupRxLow
- FrIf/FrIfConfig/FrIfCluster/FrIfGdWakeupRxWindow
- FrIf/FrIfConfig/FrIfCluster/FrIfGdWakeupTxActive
- FrIf/FrIfConfig/FrIfCluster/FrIfGdWakeupTxIdle
- FrIf/FrIfConfig/FrIfCluster/FrIfJobList/FrIfAbsTimerRef
- FrIf/FrIfConfig/FrIfCluster/FrIfJobList/FrIfJob/FrIfCommunicationOperation/FrIfRxComOpMaxLoop
- FrIf/FrIfConfig/FrIfCluster/FrIfSafetyMargin
- FrIf/FrIfConfig/FrIfFrameStructure/FrIfByteOrder
- FrIf/FrIfConfig/FrIfMaxPduCnt
- FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfRxPdu/FrIfUserRxIndicationUL
- FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfTxPdu/FrIfUserTxUL
- FrIf/FrIfGeneral/FrIfBusMirroringSupport
- FrIf/FrIfGeneral/FrIfCancelTransmitSupport
- FrIf/FrIfGeneral/FrIfDisableTransceiverBranchSupport
- FrIf/FrIfGeneral/FrIfEnableTransceiverBranchSupport
- FrIf/FrIfGeneral/FrIfFreeOpAApiName
- FrIf/FrIfGeneral/FrIfFreeOpBApiName
- FrIf/FrIfGeneral/FrIfFreeOpsHeader
- FrIf/FrIfGeneral/FrIfGetTransceiverErrorSupport
- FrIf/FrIfGeneral/FrIfMaxPduCnt
- FrIf/FrIfGeneral/FrIfReadCCConfigApi

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfAbsTimer/FrIfAb-

sTimerIdxFrIfAbsTimerIdx

[RTA-BSW] Parent Container	FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfAbsTimer/FrIfAbsTimerIdx
[AUTOSAR] Parent Container	FrIf/FrIfGeneral/FrIfAbsTimerIdx
[RTA-BSW] Description	This parameter provides a zero-based consecutive index of the absolute timers. Upper layer BSW modules use this index to identify an absolute timer.
[AUTOSAR] Description	Maximum number of supported absolute timers.
[REMOVED] Maximum Value	15
[RTA-BSW] Minimum Value	0
[AUTOSAR] Minimum Value	1

1.6.1.2 FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfCycleRepetition

[RTA-BSW] Introduction	Possible values for FlexRay Protocol version 2.1: 1,2,4,8,16,32,64
[AUTOSAR] Introduction	Possible values for FlexRay Protocol version 2.1: 1,2,4,8,16,32,64 Possible values for FlexRay Protocol version 3.0: 1,2,4,5,8,10,16, 20,32,40,50,64

1.6.1.3 FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfNoneMode

[RTA-BSW] Parent Container	FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfNoneMode
[AUTOSAR] Parent Container	FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfTxPdu/FrIfNoneMode
[RTA-BSW] Description	Using the "None-Mode" which means that there is no API FrIf_Transmit call of the upper layer for this PDU.
[AUTOSAR] Description	Using the “None-Mode” which means that there is no API FrIf_Transmit call of the upper layer for this PDU.
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[REMOVED] Default Value	false

1.6.1.4 FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfRe-

configurableFrIfReconfigurable

[RTA-BSW] Parent Container	FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfReconfigurable
[AUTOSAR] Parent Container	FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfLPdu/FrIfReconfigurable
[RTA-BSW] Description	Enable/Disable reconfiguration and disabling of this LPdu during runtime via FrIf_ReconfigLPdu() and FrIf_DisableLPdu() APIs repectively.
[AUTOSAR] Description	This parameter specifies that this LPdu is reconfigurable using FrIf_ReconfigLPdu. This means that this LPdu can be assigned to a different FrameTriggering at runtime. However, this reconfiguration is limited by hardware constraints. The direction of the LPdu cannot be reconfigured.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	false

1.6.1.5 FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfLPdu/FrIfReconfigurable

[RTA-BSW] Description	This parameter specifies that this LPdu is reconfigurable using FrIf_ReconfigLPdu. This means that this LPdu can be assigned to a different FrameTriggering at runtime. However, this reconfiguration is limited by hardware constraints. The direction of the LPdu cannot be reconfigured. Since this parameter is not used, the lower multiplicity value of this parameter is changed to 0. This is to remove the BCT validation warnings. This parameter is not related to FrIf_ReconfigLPdu() and/or FrIf_DisableLPdu() APIs. To enable this LPdu to be used with FrIf_ReconfigLPdu() and/or FrIf_DisableLPdu() APIs, refer FrIfReconfigurable parameter under FrIfFrameTriggering container.
[AUTOSAR] Description	This parameter specifies that this LPdu is reconfigurable using FrIf_ReconfigLPdu. This means that this LPdu can be assigned to a different FrameTriggering at runtime. However, this reconfiguration is limited by hardware constraints. The direction of the LPdu cannot be reconfigured.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.6 Frlf/FrlfConfig/FrlfCluster/FrlfController/FrlfTransceiver

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.7 Frlf/FrlfConfig/FrlfCluster/FrlfGChannels

[RTA-BSW] Description	The channels that are used by the cluster. Implementation Type: Fr_ChannelType
[AUTOSAR] Description	The channels that are used by the cluster.

1.6.1.8 Frlf/FrlfConfig/FrlfCluster/FrlfGMacroPerCycle

[RTA-BSW] Description	Number of macroticks in a communication cycle. Note: Lower limit 10 for FlexRay Protocol 2.1 Rev. A compliance
[AUTOSAR] Description	Number of macroticks in a communication cycle.

1.6.1.9 Frlf/FrlfConfig/FrlfCluster/FrlfGNumberOfMinislots

[RTA-BSW] Description	Number of minislots in the dynamic segment. Remark: Upper limit 7986 for FlexRay Protocol 2.1 Rev. A compliance
[AUTOSAR] Description	Number of minislots in the dynamic segment

1.6.1.10 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrlfJob/FrlfCommunication-Operation

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.11 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrlfJob/FrlfMaxIsrDelay

[RTA-BSW] Description	The maximum delay in macroticks the Frlf_JoblistExec_←cluster›() function is processed after the absolute timer interrupt was triggered. Since this parameter is not supported, the lower multiplicity value of this parameter is changed to 0. This is to remove the BCT validation warnings.
[AUTOSAR] Description	The maximum delay in macroticks the Frlf_JobListExec_‹FrlfCluster.ShortName›() function is processed after the absolute timer interrupt was triggered.
[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

1.6.1.12 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfBaseCycle

[RTA-BSW] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfBaseCycle
[AUTOSAR] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfController/FrlfFrameTriggering/FrlfBaseCycle
[ADDED] Default Value	0

1.6.1.13 Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfCycleRepetition

[RTA-BSW] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrifJobListWizard/FrlfJobEntry/FrlfCycleRepetition
[AUTOSAR] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfController/FrlfFrameTriggering/FrlfCycleRepetition
[RTA-BSW] Description	This parameter contains the FlexRay Cycle Repetition used to transmit this FlexRay Frame.possible Values: 1,2,4,8,16,32,64.
[AUTOSAR] Description	This parameter contains the FlexRay Cycle Repetition used to transmit this FlexRay Frame.
[REMOVED] Introduction	Possible values for FlexRay Protocol version 2.1: 1,2,4,8,16,32,64 Possible values for FlexRay Protocol version 3.0: 1,2,4,5,8,10,16,20,32,40,50,64
[ADDED] Default Value	1

1.6.1.14 Frlf/FrlfConfig/FrlfCluster/FrlfMaxIsrDelay

[RTA-BSW] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfMaxIsrDelay
[AUTOSAR] Parent Container	Frlf/FrlfConfig/FrlfCluster/FrlfJobList/FrlfJob/FrlfMaxIsrDelay
[RTA-BSW] Description	The maximum delay in macroticks the Frlf_JoblistExec_<cluster>() function is processed after the absolute timer interrupt was triggered.
[AUTOSAR] Description	The maximum delay in macroticks the Frlf_JobListExec_<FrlfCluster.ShortName>() function is processed after the absolute timer interrupt was triggered.

1.6.1.15 Frlf/FrlfConfig/FrlfFrameStructure/FrlfPdusInFrame/FrlfPduUpdateBitOffset

[RTA-BSW] Minimum Value	-1

[AUTOSAR] Minimum Value	0
[ADDED] Default Value	-1

1.6.1.16 FrIf/FrIfConfig/FrIfPdu/FrIfPduDirection/FrIfTxPdu/FrIfNoneMode

[RTA-BSW] Upper Multiplicity	0
[AUTOSAR] Upper Multiplicity	1

1.6.1.17 FrIf/FrIfGeneral/FrIfAbsTimerIdx

[RTA-BSW] Description	Maximum number of supported absolute timers. Since this parameter is not supported, the lower multiplicity value of this parameter is changed to 0. This is to remove the BCT validation warnings.
[AUTOSAR] Description	Maximum number of supported absolute timers.
[RTA-BSW] Multiplicity	0..0
[AUTOSAR] Multiplicity	1..1

1.7 API Definition

List of supported API functions

Table 1.56.FrIf Supported API functions

`FrIf_Init`	Initialises the FlexRay interface
`FrIf_ControllerInit`	Initialises a FlexRay Communication Controller (CC)
`FrIf_SetAbsoluteTimer`	A wrapper for the FlexRay Driver API function `Fr_SetAbsoluteTimer`
`FrIf_EnableAbsoluteTimerIRQ`	A wrapper for the FlexRay Driver API function `Fr_EnableAbsoluteTimerIRQ`
`FrIf_StartCommunication`	A wrapper for the FlexRay Driver API function `Fr_StartCommunication`
`FrIf_HaltCommunication`	A wrapper for the FlexRay Driver API function `Fr_HaltCommunication`
`FrIf_AbortCommunication`	A wrapper for the FlexRay Driver API function `Fr_AbortCommunication`
`FrIf_GetState`	Returns the current FlexRay interface state
`FrIf_SetState`	Requests a transition to another FlexRay interface state
`FrIf_SetWakeupChannel`	A wrapper for the FlexRay Driver API function `Fr_SetWakeupChannel`
`FrIf_SendWUP`	A wrapper for the FlexRay Driver API function `Fr_SendWUP`
`FrIf_GetPOCStatus`	A wrapper for the FlexRay Driver API function `Fr_GetPOCStatus`

FrIf_GetGlobalTime	A wrapper for the FlexRay Driver API function Fr_GetGlobalTime
FrIf_AllowColdStart	A wrapper for the FlexRay Driver API function Fr_AllowColdStart
FrIf_GetMacroticksPerCycle	Returns the number of Macroticks per cycle
FrIf_GetMacrotickDuration	Returns the duration of a Macrotick in nanoseconds
FrIf_Transmit	Requests the transmission of a PDU (Protocol Data Unit)
FrIf_SetTransceiverMode	A wrapper for the FlexRay Driver API function FrTrcv_SetTransceiverMode
FrIf_GetTransceiverMode	A wrapper for the FlexRay Driver API function FrTrcv_GetTransceiverMode
FrIf_GetTransceiverWUReason	A wrapper for the FlexRay Driver API function FrTrcv_GetTransceiverWUReason
FrIf_ClearTransceiverWakeup	A wrapper for the FlexRay Driver API function FrTrcv_ClearTransceiverWakeup
FrIf_CancelAbsoluteTimer	A wrapper for the FlexRay Driver API function Fr_CancelAbsoluteTimer
FrIf_DisableAbsoluteTimerIRQ	A wrapper for the FlexRay Driver API function Fr_DisableAbsoluteTimerIRQ
FrIf_GetCycleLength	Returns the configured time of the GdCycle parameter (in nanoseconds) for the specified FlexRay Communication Controller
FrIf_GetVersionInfo	Returns version information for this module
FrIf_GetNmVector	A wrapper for the FlexRay Driver API function Fr_GetNmVector
FrIf_GetWakeupRxStatus	A wrapper for the FlexRay Driver API function Fr_GetWakeupRxStatus
FrIf_GetChannelStatus	A wrapper for the FlexRay Driver API function Fr_GetChannelStatus
FrIf_GetClockCorrection	A wrapper for the FlexRay Driver API function Fr_GetClockCorrection
FrIf_GetNumOfStartupFrames	A wrapper for the FlexRay Driver API function Fr_GetNumOfStartupFrames
FrIf_AllSlots	A wrapper for the FlexRay Driver API function Fr_AllSlots
FrIf_ReadCCConfig	A wrapper for the FlexRay Driver API function Fr_ReadCCConfig
FrIf_ReconfigLPdu	A wrapper for the FlexRay Driver API function Fr_ReconfigLPdu
FrIf_DisableLPdu	A wrapper for the FlexRay Driver API function Fr_DisableLPdu
FrIf_GetSyncFrameList	A wrapper for the FlexRay Driver API function Fr_GetSyncFrameList
FrIf_MainFunction	The central processing function of FrIf

For Further information please refer to AUTOSAR_SWS_FlexRayInterface.pdf

1.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

1.8.1 Reconfigure or disable L-PDU

To enable reconfiguring of L-PDUs (using the API function call FrIf_ReconfigLPdu), set the following configuration parameter values:

- FrIf/FrIfGeneral/FrIfReconfigLPduSupport = TRUE
- FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfReconfigurable = TRUE

To enable the disabling of L-PDUs (using the API function call FrIf_DisableLPdu), set the following configuration parameter values:

- FrIf/FrIfGeneral/FrIfReconfigLPduSupport = TRUE
- FrIf/FrIfGeneral/FrIfDisableLPduSupport = TRUE
- FrIf/FrIfConfig/FrIfCluster/FrIfController/FrIfFrameTriggering/FrIfReconfigurable = TRUE

Note that only reconfigurable L-PDUs can be disabled.

1.8.2 General Advice

- FrIf supports only one cluster and one controller.
- A minimum of one Tx and one Rx FrIf PDU needs to be configured.
- The configuration parameter FrIfVendor11Driver should be set to FALSE.
- If you wish to disable the FRIF_FILL_BYTES feature, you should not configure a value for the FrIfUnusedBitValue parameter (and uncheck the default).
- If you set FrIfZeroPayloadTx to TRUE, then you must also configure FrIfUnusedBitValue parameter.
- The short name of all FrIfFrameTriggerings should be unique during dual-controller configuration.
- The purpose of using the specific parameter FrIfJoblistExecDelayCompensation is to compensate the delay between Fr_GetGlobalTime API and Fr_SetAbsoluteTimer API, called within FrIf_JobListExec_0 API. The delay between these two APIs can vary from project to project. Therefore only

configure this parameter only when required.

- In order to support backward compatibility of AUTOSAR in FrIf, the configuration parameter FrIfEnableARBackwardCompatibility should be set to TRUE.
- The parameter FrIfTxPduId should not be configured manually by the user.
- There are two template files available for the GeneralTypes definitions from Fr, FrTrcv modules:

Fr_GeneralTypes.h_tpl and Fr_GeneralTypes.h_ETAS_tpl. One or the other need to be selected for integration depending on the driver use case:

- Fr_GeneralTypes.h_tpl contains the type definitions compliant with the latest version of Autosar. This is the file to use by default for RTA-CAR integration.
- Fr_GeneralTypes.h_ETAS_tpl contains the type definitions with ETAS specific adaptations to support HW types not compatible with Autosar or to support memory optimization. This file therefore contains deviations from the latest version of Autosar.

1.9 Integration Advice

1.9.1 Init Functions

FrIf_Init must be called for each configured FlexRay CC during the system initialization.

During initialisation, the State of FrIf is OFFLINE and no Communication is possible. The FrIf State goes to ONLINE when FrSM goes to online.

1.9.2 DeInit Functions

When an application wants to stop the Flexray communication, FrIf_HaltCommunication is called via FrSM. The communication is halted and the FrIf State is changed to OFFLINE.

1.9.3 Scheduled Functions

When integrating FrIf, the macro FRIF_JOBLIST_MONITORING (in FrIfCfg_SchM.h) should be set to ON. This allows the main scheduling function to call FrIf_SyncJob to synchronize when FrSM State is ONLINE.

1.9.4 Configuring Fr_GeneralTypes.h

The Fr_GeneralTypes.h is now an integration file and should be configured by the User (The supported changes are in AUTOSAR and are backwards compatible up to AR 4.0.3).

1.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to

disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FrIf module uses the following OS resources:

- FrIf_StateMachineNoNest
- FrIf_IpduCntTrigNoNest
- FrIf_BufferAndCntTrigNoNest
- FrIf_PayloadNoNest
- FrIf_JobListNoNest

1.10 Hardware Requirements

Table 1.57.FrIf Hardware Requirements

Support	Usage
FLASH	22332 bytes
RAM	655 bytes

Usage calculated based on a sample configuration.

2 FlexRay State Manager

Module name	FrSM
Package	RTA-FRAY
AUTOSAR Module ID	142
AUTOSAR Version	22-11

2.1 Introduction

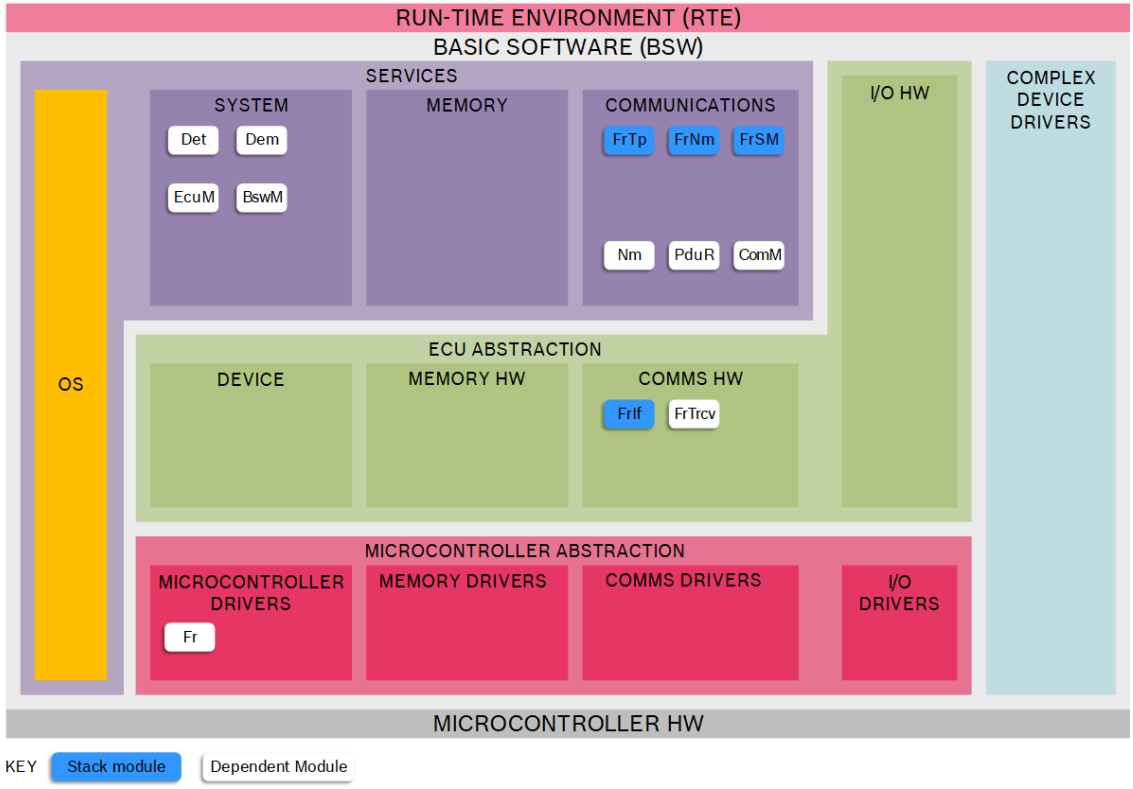


Figure 2.1.FlexRay Architecture

The FlexRay State Manager module (FrSM) is located in the communication services layer, above the communication drivers layer and communication HW abstraction layer. The diagram above shows the position of FrSM in the AUTOSAR layered architecture.

FrSM provides a hardware-abstracted mechanism to control the startup, wakeup and shutdown of nodes in a FlexRay cluster by implementing a finite state machine which maps communication modes to an equivalent bus state. It supports the detection and switching of communication modes ('active', 'passive' and 'keyslot only' modes).

FrSM recognises the following AUTOSAR-specified states:

- FRSM_READY

- FRSM_WAKEUP
- FRSM_STARTUP
- FRSM_HALT_REQ
- FRSM_ONLINE
- FRSM_ONLINEPASSIVE
- FRSM_ECU_ACTIVE
- FRSM_ECU_PASSIVE
- FRSM_KEYSLOT_ONLY

In addition to those listed above, the following additional state types are also provided:

- FRSM_UNDEFINED (The state could not be unequivocally retrieved).
- FRSM_RESET (The default state after power-on).

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

2.2 Supported Features

2.2.1 Multiple controllers

If a cluster contains an ECU with a second enabled controller, the FrSM assigns a state based on the combination of controller POC states as follows:

CC 1 State	CC 2 State	FrSM State
FR_POCSTATE_READY	FR_POCSTATE_READY	READY
FR_POCSTATE_WAKEUP	FR_POCSTATE_READY	WAKEUP
FR_POCSTATE_STARTUP	FR_POCSTATE_STARTUP	STARTUP
FR_POCSTATE_NORMAL_ACTIVE	FR_POCSTATE_STARTUP	STARTUP
FR_POCSTATE_STARTUP	FR_POCSTATE_NORMAL_ACTIVE	STARTUP
FR_POCSTATE_NORMAL_ACTIVE	FR_POCSTATE_NORMAL_ACTIVE	NORMAL_ACTIVE
FR_POCSTATE_STARTUP	FR_POCSTATE_KEYSLOT_ONLY	STARTUP
FR_POCSTATE_KEYSLOT_ONLY	FR_POCSTATE_STARTUP	STARTUP
FR_POCSTATE_KEYSLOT_ONLY	FR_POCSTATE_KEYSLOT_ONLY	KEYSLOT_ONLY
FR_POCSTATE_NORMAL_ACTIVE	FR_POCSTATE_KEYSLOT_ONLY	KEYSLOT_ONLY
FR_POCSTATE_KEYSLOT_ONLY	FR_POCSTATE_NORMAL_ACTIVE	KEYSLOT_ONLY
FR_POCSTATE_KEYSLOT_ONLY	FR_POCSTATE_PASSIVE_ONLINE	PASSIVE_ONLINE
FR_POCSTATE_PASSIVE_ONLINE	FR_POCSTATE_KEYSLOT_ONLY	PASSIVE_ONLINE
FR_POCSTATE_HALT	FR_POCSTATE_NORMAL_ACTIVE	HALT

FR_POCSTATE_NORMAL_ACTIVE	FR_POCSTATE_HALT	HALT

2.3 Extensions

2.3.1 Configuration Extensions

The following configuration parameters are provided in version v12.7.0 in addition to those defined in AR v22-11

2.3.1.1 FrSM/FrSMConfig/FrSMCluster/FrSMChannel

Short Name	FrSMChannel
Description	This parameter refers the FlexRay Transceiver channel.
Multiplicity	1..1
Default Value	FRSM_CHANNEL_B
Enumeration Literals	FRSM_CHANNEL_A, FRSM_CHANNEL_B
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.2 FrSM/FrSMConfig/FrSMCluster/FrSMClusterDemEventParameter-Refs/FrSMRbDynCreate_FRSM_E_CLUSTER_STARTUP

Short Name	FrSMRbDynCreate_FRSM_E_CLUSTER_STARTUP
Description	Selects the way how the FrSM DemEvents are created. If not set or set to “Disabled”, no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRSM_E_CLUSTER_STARTUP) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.3 FrSM/FrSMConfig/FrSMCluster/FrSMClusterDemEventParameter-Refs/FrSMRbDynCreate_FRSM_E_CLUSTER_SYNC_LOSS

Short Name	FrSMRbDynCreate_FRSM_E_CLUSTER_SYNC_LOSS

Description	Selects the way how the FrSM DemEvents are created. If not set or set to “Disabled”, no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter FRSM_E_CLUSTER_SYNC_LOSS) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.4 FrSM/FrSMGeneral/FrSMCddHeaderFile

Short Name	FrSMCddHeaderFile
Description	This parameter defines header files for callback functions which are implemented by CDD, e.g. <Cdd>_SyncLossErrorIndication.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

2.3.1.5 FrSM/FrSMGeneral/FrSMDSMReInit

Short Name	FrSMDSMReInit
Description	Enables or disables the reporting of DEM event after DSM Re-initialisation
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.6 FrSM/FrSMGeneral/FrSMInitCallBack

Short Name	FrSMInitCallBack
Description	Enables and disables the call of the init callback function
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.7 FrSM/FrSMGeneral/FrSMSystemTimerUsed

Short Name	FrSMSystemTimerUsed

Description	Enables and disables the use of system timer in the project. When it is enabled, the system timer will be used for the FrSM timers T1, T2, T3 and T4 and when disabled, the OS timer will be used for the same.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.2 Additional Services

The following API functions are provided in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API functions defined in this version.

2.4 Limitations

The following limitations apply to version v12.7.0:

- One FlexRay cluster and two controllers are supported for communication purposes.

  When two controllers are configured per ECU:
  - Those two controllers should belong to the same cluster.
  - Wakeup pattern transmission will be triggered only in CC0 (Communication Controller 0) even when two controllers are configured.
  - The FrSM_State variable will be on the cluster level and not the controller level (i.e the global variable FrSM_State will provide the current state of the cluster).

2.5 Exclusions

There are no unsupported API functions in this release.

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FrSM/FrSMConfig/FrSMCluster/FrSMMinNumberOfColdstarter

2.6 Deviations

2.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

2.6.1.1 FrSM/FrSMConfig/FrSMCluster/FrSMClusterDemEventParameter-Refs/FRSM_E_CLUSTER_STARTUP

[RTA-BSW] Lower Multiplicity	0

[AUTOSAR] Lower Multiplicity	1

2.6.1.2 FrSM/FrSMConfig/FrSMCluster/FrSMClusterDemEventParameter-Refs/FRSM_E_CLUSTER_SYNC_LOSS

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

2.7 API Definition

List of supported API functions

Table 2.12.FrSM Supported API functions

`FrSM_Init`	Initializes the FlexRay State Manager
`FrSM_RequestComMode`	Used by the ComM module to startup or shutdown the communication on a FlexRay cluster
`FrSM_GetCurrentComMode`	Used to determine the current communication mode of a FlexRay cluster
`FrSM_GetVersionInfo`	Returns the version information of the module
`FrSM_AllSlots`	Used to leave the KeySlotOnlyMode
`FrSM_SetEcuPassive`	Set the state of all FrSM state machines to passive (or active)

For Further information please refer to AUTOSAR_SWS_FlexRayStateManager.pdf

2.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

2.8.1 Interfaces

The FrSM module uses an internal function `SyncLossErrorIndication` to detect when an ECU controlled by a specific CDD (Complex Device Driver) loses sync with the FlexRay cluster. You can configure this function to use a different name by setting the `FrSMSyncLossErrorIndicationName` configuration value. Note that if you use this feature, you must also define a header file for the callback functions implemented by the CDD (see 'Callback functions' below).

2.8.2 Timers

Configure Timer 1 by entering a value (>0) in the attribute `FrSmDurationT1` present in the container `FrSmCluster`. Timer 1 represents the time between a wakeup pattern being sent and the sending of a coldstart sequence, so you cannot configure Timer 1 if a node is not a wakeup node.

Configure Timer 2 by entering a value (>0) in the attribute `FrSmDurationT2` present in the container `FrSmCluster`. Timer 2 represents the duration of restart or resynchronization attempts for a CC within the cluster. If Timer 2 is configured, the fields `StartupRepetitions`, `WakeupRepetitions` and `StartupRepetitionsWithWakeup` are generated in the FrSm configuration structure.

2.8.3 Wakeup & Cold Start

To configure a node in the cluster as a wakeup node, set the attribute `FrSmIsWakeupEcu` to ON. Note that if the wakeup feature is not configured, Timer 1 configuration is set to OFF and the duration is not generated in the module configuration structure.

To configure a node in the cluster as a cold start node, set the attribute `FrSmIsColdstartEcu` to ON. If cold start is not configured for a node, then it does not become a leading/following cold start node. Note that you should also configure the FrIf module for cold start if using this feature.

2.8.4 Callback Functions

If you've used the `FrSMSyncLossErrorIndicationName` configuration value to specify an alternative function for detecting loss of sync for a CDD-controlled ECU, you must set the configuration value `FrSMCddHeaderFile` to the name of the header file which contains the definitions for the function. Note that this should be the name of the header file descriptor only, e.g. 'Cdd' and not 'Cdd.h'.

You can set the configuration value `FrSMInitCallBack` to TRUE if you wish to define a function to be called after initialisation (this can enable faster startup of the FlexRay cluster). The file FrSM_Cfg_Callback.c provides example code for such a callback function.

2.8.5 Timer Configuration

You can set the FrSM module to use either the system timer or OS timer by setting the configuration value `FrSMSystemTimerUsed` to TRUE or FALSE respectively.

Timer t4 indicates the wait time before the FrSM module re-initialises the controllers on a cluster and sets the transceiver mode to 'standby'. To use a set value for this delay instead of timer t4, set the configuration value `FrSMTrcvStdbyDelay` present in the container `FrSmCluster` to a value in seconds (>0).

As the starting of this timer is not in sync with the main function cycle, while generating the count value for the configured duration, 1 is added to it. For example, if the configured value is 1 sec and the main function cycle is 5ms, then the

generated count value will be 201 (1000/5 + 1 = 201) counts.

2.8.6 Key-Slot Mode

The `FrSMAllSlotsSupport` configuration parameter can be used to enable/disable FrSM support to switch the key-slot/single-slot mode to all-slot mode, using TRUE or FALSE respectively.

2.8.7 DEM Event Reporting

Reporting of the DEM event after DSM re-initialisation can be enabled or disabled using `FrSMDSMReInit`, and setting to TRUE or FALSE respectively.

2.9 Integration Advice

2.9.1 Init Functions

FrSM_Init initializes the FlexRay State Manager.

On initialisation the FrSM module, all communication controllers in the associated cluster have their states set to FRSM_READY.

2.9.2 DeInit Functions

No DeInit is required.

The state FRSM_HALT_REQ is the shutdown state of FrSM, where all internal processes within the communication controller are stopped. This state is reached when an internal error of the CC was found and transition to FRSM_READY is made.

2.9.3 Scheduled Functions

The FrSM module processes the state machine for a cluster via FrSm_MainFunction, which should be scheduled cyclically by SchM with a cycle time that is shorter than or equal to the FlexRay cycle duration.

The FrSM timers can be configured using the system timer or the OS timer. A macro `FRSM_SYSTEM_TIMER_USED` is provided (in FrSM_Cfg.h) which can enable or disable the use of the system timer.

The timers Timer1, Timer2, Timer3 and Timer4 are implemented in FrSM as software timers (see the Configuration Advice). If the Timers are running, they are incremented in FrSM_MainFunction_0, which should be scheduled once per CC cycle. Hence the timers will only work properly when FrSM_MainFunction_0 is scheduled properly

To use the system timer for running the FrSM timers, you should alter the file FrSM_Cfg_Time.h, and then enable the timer using the macro described above. Sample code is included in the .h file for reference. The FrSM timers will then run on the system timer ticks.

To use the internal software timers, you should use the macro described above to disable the system timer, and then ensure that the FrSM main function is correctly scheduled. This will then increment the software timers during normal operation.

2.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FrSM module uses the following OS resources:

- FrSM_ReqstModeNoNest
- FrSM_StateNoNest
- FrSM_EcuPassiveNoNest

2.10 Hardware Requirements

Table 2.13.FrSM Hardware Requirements

Support	Usage
FLASH	4932 bytes
RAM	94 bytes

Usage calculated based on a sample configuration.

3 FlexRay Network Management

Module name	FrNm
Package	RTA-FRAY
AUTOSAR Module ID	32
AUTOSAR Version	22-11

3.1 Introduction

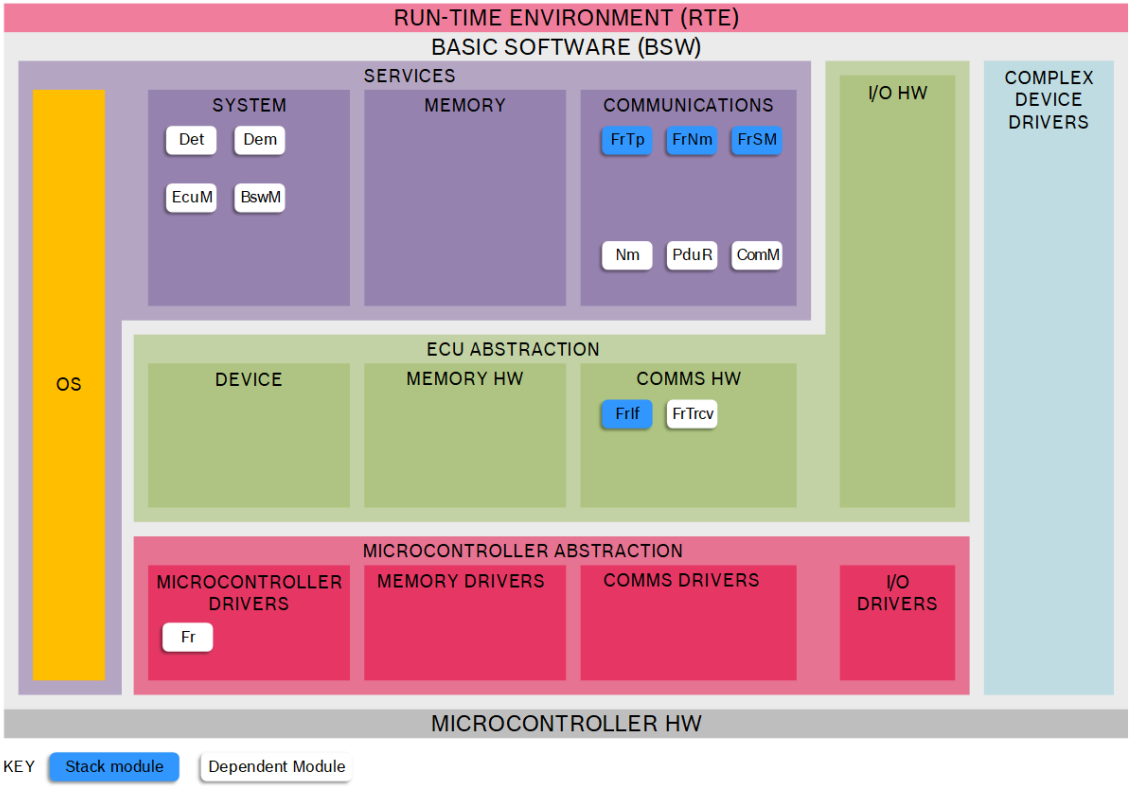


Figure 3.1.FlexRay Architecture

The FlexRay Network Management module (FrNm) is located in the communication services layer, above the communication drivers layer and communication HW abstraction layer. The diagram above shows the position of FrNm in the AUTOSAR layered architecture.

FrNm provides a hardware-abstracted mechanism to co-ordinate the transition between normal operation and bus-sleep mode of the FlexRay network. Application API function calls requesting network access are passed to Nm from the Communication Manager (ComM) for all FlexRay clusters, via the Network Management Interface (Nm).

FrNm interacts with the FlexRay Interface module (FrIf) for the transmission/reception of PDUs, and it implements a state machine which signals the bus/network states and modes to/from the Nm module.

The FrNm module recognises the following network states:

- NM_STATE_UNINIT
- NM_STATE_BUS_SLEEP
- NM_STATE_PREPARE_BUS_SLEEP
- NM_STATE_READY_SLEEP
- NM_STATE_NORMAL_OPERATION
- NM_STATE_REPEAT_MESSAGE
- NM_STATE_SYNCHRONIZE

The FrNm module recognises the following network modes:

- NM_MODE_BUS_SLEEP
- NM_MODE_PREPARE_BUS_SLEEP
- NM_MODE_SYNCHRONIZE
- NM_MODE_NETWORK

Module Configuration

This module can be generated by the RTA-BSW Configuration Generator.

For any unsupported automatic configuration, the module can be manually configured by the user.

3.2 Supported Features

3.2.1 Transmission/Reception

FrNm supports both single and multiple PDUs when scheduling transmission and reception operations. It is able to distinguish between PDUs in static and dynamic segments, and between NM-Vote and NM-Data PDUs. Combined PDUs (NM-Vote and NM-Data) are also supported. The setting of CBV (Control Bit Vector) in NM-Data and combined PDUs is also supported.

3.2.2 Synchronisation

FrNm is able to recognise both short-term and long-term synchronisation loss events, and set the state of the target FlexRay cluster accordingly.

3.2.3 Partial Networking

FrNm supports PNC (Partial Network Cluster) operations as per the specification of the AUTOSAR ComM module. This feature may be used to reduce the power consumption of communication domains. However, you should note that older ECUs may not support PNC operations.

3.2.4 User Data

FrNm supports the setting and retrieval of user data via the Com module. See the FrNmComUserDataSupport configuration parameter in the Configuration Advice.

3.2.5 Dual Channel PDU

FrNm supports the transmission and reception of PDUs on dual FlexRay channels (A and B) by sending and receiving two PDUs, one for FlexRay channel A, and one for FlexRay channel B. See the FrNmDualChannelPduEnable configuration parameter in the Configuration Advice.

3.2.6 Configuration Class

FrNm supports 2 kinds of configuration class -

- Pre-compile
- Post-build Loadable

FrNm also supports Post Build Selectable for the PduR component. The validation while supporting PduR for PBS is provided for only the first value of the parameters accessed from PduR.

See the Configuration Advice for more details.

3.3 Extensions

3.3.1 Configuration Extensions

The following configuration parameters are provided in version v12.7.0 in addition to those defined in AR v22-11

3.3.1.1 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmPnEraCalcEnabled

Short Name	FrNmPnEraCalcEnabled
Description	Specifies if FrNm calculates the PN request information for external requests. (ERA)
Introduction	false: PN request are not calculated true: PN request are calculated
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.2 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmPnEraRxNSduRef

Short Name	FrNmPnEraRxNSduRef
Description	Reference to a global Pdu. The SduRef is required for every FrNm Channel, because ERA is reported per channel.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.3 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/IsFrNmMainFuncBeSchedulable

Short Name	IsFrNmMainFuncBeSchedulable
Description	This parameter is used for scheduling the FrNm Main function. TRUE:FrNm_MainFunction will be available as a schedulable entity in Bswmd. FALSE:FrNm_MainFunction will not be available as a schedulable entity in Bswmd
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.4 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/IsFrNmMainFuncTxBeSchedulable

Short Name	IsFrNmMainFuncTxBeSchedulable
Description	This parameter is used scheduling the FrNm Tx function. TRUE:FrNm_MainFunctionTx API will be available as a schedulable entity in Bswmd. FALSE:FrNm_MainFunctionTx API will not be available as a schedulable entity in Bswmd
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.5 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelTiming/FrNmMainFunctionTxPeriod

Short Name	FrNmMainFunctionTxPeriod
Description	This parameter defines the processing cycle of the main functionTx of FrNm module in seconds.
Multiplicity	0..1
Value Range	0..INF
Default Value	0.005
Type	ECUC-FLOAT-PARAM-DEF

3.3.1.6 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelTiming/FrNmMsgTimeoutTime

Short Name	FrNmMsgTimeoutTime

Description	Timeout of a NM-message. It determines in seconds how long the NM shall wait with notification of transmission failure while communication errors occur on the bus.
Multiplicity	0..1
Value Range	0..65.535
Default Value	0
Type	ECUC-FLOAT-PARAM-DEF

3.3.1.7 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmComControlEnabled

Short Name	FrNmComControlEnabled
Description	Pre-processor switch for enabling i14229 Communication Control support.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.8 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnEiraCalcEnabled

Short Name	FrNmPnEiraCalcEnabled
Description	Specifies if FrNm calculates the PN request information for internal an external requests. (EIRA)
Introduction	true: PN request are calculated false: PN request are not calculated
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.9 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnEiraRxNSduRef

Short Name	FrNmPnEiraRxNSduRef
Description	Reference to a Pdu in the COM-Stack.
Introduction	Only one SduRef is required for FrNm because the EIRA is the aggregation over all FlexRay Channels.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

3.3.1.10 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo

Short Name	FrNmPnInfo
Description	PN information configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.11 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNm-MaxNumberOfPnsConfigurableAtPostBuildTime

Short Name	FrNmMaxNumberOfPnsConfigurableAtPostBuildTime
Description	Specifies the maximum number of the PNs can be configured during post build time.This parameter should be configured during pre compile time itself.
Multiplicity	0..1
Value Range	0..56
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.12 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNmP-nFilterMaskByte

Short Name	FrNmPnFilterMaskByte
Description	Filter mask byte configuration
Multiplicity	0..7
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.13 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNmP-nFilterMaskByte/FrNmPnFilterMaskByteIndex

Short Name	FrNmPnFilterMaskByteIndex
Description	Index of the filter mask byte. Specifies the position within the filter mask byte array.
Multiplicity	1..1
Value Range	0..6
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.14 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNmPnFilterMaskByte/FrNmPnFilterMaskByteValue

Short Name	FrNmPnFilterMaskByteValue
Description	Parameter to configure the filter mask byte.
Multiplicity	1..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.15 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNmPnInfoLength

Short Name	FrNmPnInfoLength
Description	Specifies the length of the PN request information in the NM message.
Multiplicity	1..1
Value Range	1..7
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.16 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnInfo/FrNmPnInfoOffset

Short Name	FrNmPnInfoOffset
Description	Specifies the offset of the PN request information in the NM message.
Multiplicity	1..1
Value Range	1..7
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

3.3.1.17 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnResetTime

Short Name	FrNmPnResetTime
Description	Specifies the runtime of the reset timer in seconds. This reset time is valid for the reset of PN requests in the EIRA and in the ERA. The value shall be the same for every channel. Thus it is a global config parameter.
Multiplicity	0..1
Value Range	0.001..65.535
Default Value	-

Type	ECUC-FLOAT-PARAM-DEF

3.3.1.18 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmPnSyncShutdownErrorReactionEnabled

Short Name	FrNmPnSyncShutdownErrorReactionEnabled
Description	Pre-processor switch for enabling reaction, if a top-level PNC coordinator received a PN shutdown message on a NM-channel which refer to a ComM channel that is actively coordinated by a PNC gateway.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.2 Additional Services

The following API functions are provided in version v12.7.0 in addition to those defined in AR v22-11

3.3.2.1 FrNm_RequestSynchronizedPncShutdown

Syntax	`FrNm_RequestSynchronizedPncShutdown(NetworkHandleType NetworkHandle, PNCHandleType pncId)`
Parameters In	`NetworkHandle` - Identification of the NM-channel `pncId`
Return Type	`E_OK` - No error `E_NOT_OK` - PNC shutdown Request has failed
Description	Causes a PNC shutdown request to be transmitted next on the bus.

3.4 Limitations

The following features are not supported for version v12.7.0:

- Only one FlexRay cluster is supported in this release, therefore the FrNm module only supports a network consisting of a single cluster.

3.5 Exclusions

There are no unsupported API functions in this release.

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmCarWakeUpBitPosition
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmCarWakeUpBytePosition
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmCarWakeUpFilterEnabled

- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmCarWakeUpFilterNodeId
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmCarWakeUpRxEnabled
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmDynamicPncToChannelMappingEnabled
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmRepeatMsgIndEnabled
- FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelTiming/FrNmVoteInhibitionEnabled
- FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmDynamicPncToChannelMappingSupport

3.6 Deviations

3.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

3.6.1.1 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmSynchronizedPncShutdownEnabled

[RTA-BSW] Description	This parameter defines if FrNm handle PN shutdown messages to support a synchronized PNC shutdown across a PN topology. This is only used for ECUs in the role of a top-level PNC coordinator or intermediate PNC coordinator. Thus, the PNC gateway functionality is enabled and therefore ERA calculation is used.
[AUTOSAR] Description	Specifies if FrNm handle PN shutdown messages to support a synchronized PNC shutdown across a PN topology. This is only used for ECUs in the role of a top-level PNC coordinator or intermediate PNC coordinator. Thus, the PNC gateway functionality is enabled and therefore ERA calculation is used.
[REMOVED] Introduction	FALSE: synchronized PNC shutdown is disabled TRUE: synchronized PNC shutdown is enabled.

3.6.1.2 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelIdentifiers/FrNmTxPdu/FrNmTxConfirmationPduId

[RTA-BSW] Minimum Value	-1
[AUTOSAR] Minimum Value	0

3.6.1.3 FrNm/FrNmChannelConfig/FrNmChannel/FrNmChannelTiming/FrNmReadySleepCnt

[RTA-BSW] Minimum Value	1
[AUTOSAR] Minimum Value	0

3.6.1.4 FrNm/FrNmGlobalConfig/FrNmGlobalFeatures/FrNmStateChangeIndicationEnabled

[RTA-BSW] Description	Pre-processor switch for enabling state change indication.(It is advised to set it as TRUE)
[AUTOSAR] Description	Pre-processor switch for enabling state change indication.

3.7 API Definition

List of supported API functions

Table 3.25.FrNM Supported API functions

`FrNm_Init`	Initialises the FrNm module and the network state machine
`FrNm_PassiveStartUp`	Triggers the FrNm module passive startup (switching modes from NM_MODE_BUS_SLEEP to NM_MODE_NETWORK)
`FrNm_NetworkRequest`	Requests network reservation for a communication operation
`FrNm_NetworkRelease`	Releases network reservation after a communication operation completes
`FrNm_GetState`	Requests the current state and mode of the network state machine
`FrNm_SetUserData`	Sets the user data to be included with the next transmitted network message
`FrNm_GetUserData`	Returns the user data that was included in the last transmitted network message
`FrNm_GetPduData`	Returns the PDU data included in the last received network message
`FrNm_GetNodeIdentifier`	Returns the node identifier from the last received network message
`FrNm_GetLocalNodeIdentifier`	Returns the node identifier configured for the local host node
`FrNm_RepeatMessageRequest`	Triggers a 'repeat message' request to be transmitted
`FrNm_CheckRemoteSleepIndication`	Checks whether or not remote sleep indication has taken place

`FrNm_EnableCommunication`	Enables PDU transmission on the network
`FrNm_DisableCommunication`	Disables PDU transmission on the network
`FrNm_RequestBusSynchronization`	A dummy function which has no effect on the FlexRay bus but is included for possible future development
`FrNm_Transmit`	A dummy function which has no effect on the FlexRay bus but is included for standardisation purposes
`FrNm_GetVersionInfo`	Returns version information for the module
`FrNm_RxIndication`	Called by the FrIf module when a PDU is received to acknowledge reception
`FrNm_SetSleepReadyBit`	Sets (or resets) the NM Coordinator Sleep Ready bit in the Control Bit Vector
`FrNm_StartupError`	Used to generate an error when synchronisation of the FlexRay cluster fails
`FrNm_TriggerTransmit`	Called by the FrIf module when a PDU is transmitted (triggered)
`FrNm_TxConfirmation`	Called by the FrIf module after a PDU has been transmitted to acknowledge the transmit operation

For Further information please refer to AUTOSAR_SWS_FlexRayNetworkManagement.pdf

3.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

3.8.1 Communication Control

Under normal circumstances, the FrNm_EnableCommunication and FrNm_DisableCommunication API functions are used to enable or disable PDU transmission on the network. The ability to do this enabling/disabling can itself be disabled by setting FrNmComControlEnabled to FALSE.

3.8.2 User Data (via Com)

To enable support for the setting and retrieval of user data via the Com module, set FrNmComUserDataEnabled to TRUE. If you are using this feature, note that errors/warnings will occur if:

- FrNmComUserDataSupport is set to TRUE and FrNmUserDataTxPdu is not

configured.

- FrNmComUserDataSupport is set to FALSE and FrNmUserDataTxPdu is configured.
- FrNmComUserDataSupport is set to TRUE and FrNmTxUserDataPduRef is not configured.
- The length of the I-PDU referred to by FrNmTxUserDataPduRef does not match the length of the user data.
- Both FrNmPassiveModeEnabled and FrNmComUserDataSupport are set to TRUE.

3.8.3 Dual Channel PDU

To enable the transmission and reception of PDUs on dual FlexRay channels (A and B), set FrNmDualChannelPduEnable to TRUE.

3.8.4 Partial Networking

To enable PNC (Partial Network Cluster) transmit/receive operations, set FrNmPnEnabled to TRUE. This causes FrNm to recognise and process the PN information in the message by setting the CRI bit of the CBV (Control Bit Vector) to 1. If FrNmPnEnabled is set to FALSE, the CRI bit is set to 0 in transmitted messages, indicating that no PN information is present.

If you enable PNC operations, you must consider the aggregation of PN requests. The activity of a partial network may cause an ECU to switch I-PDU groups (for example, to prevent false network timeouts). This switching is performed by the ComM module, as the FrNm module only provides information on whether or not a PN operation has been requested. As the ComM module requires PN requests to be aggregated before release, it supports both External and Internal Requests Array (EIRA) and External Request Array (ERA) aggregation types. The FrNm module supports these aggregation types using the following configuration parameters.

For EIRA aggregation:

- Set FrNmPnEiraCalcEnabled to TRUE if FrNm has to process PN request information for both internal and external requests.
- Set FrNmPnResetTime to the value that represents the run time of the reset timer. This value should be set greater than the value set for FrNmMsgCycleTime and less than the value of FrNmTimeoutTime.
- Set FrNmPnEiraRxNSduRef to refer to a PDU in the COM stack (only one reference is required for all channels).

For ERA aggregation:

- Set FrNmPnEraCalcEnabled to TRUE if FrNm has to process PN request information for external requests.
- Set FrNmPnResetTime to the value that represents the run time of the reset timer. This value should be set greater than the value set for FrNmMsgCy-

cleTimeand less than the value of FrNmTimeoutTime.

- Set FrNmPnEnabled to TRUE.
- Set FrNmPnEiraRxNSduRef to refer to a PDU in the COM stack (one reference is required for each channel).

Note that both of these methods require configuration of a PDU in ComM with the same length as that which is specified in the FrNmPnInfoLength configuration parameter. For more information on EIRA and ERA configuration, please refer to the Configuration Advice section for the ComM module.

3.8.5 Post Build Configuration

FrNm supports 2 kinds of configuration class -

- Pre-compile
- Post-build Loadable

If the Post-Build Loadable option is to be used for FrNm, the following two configuration steps must first be completed.

Step 1 Select the Configuration Class

Go to the FrNm module configuration, and then the “Properties” tab.

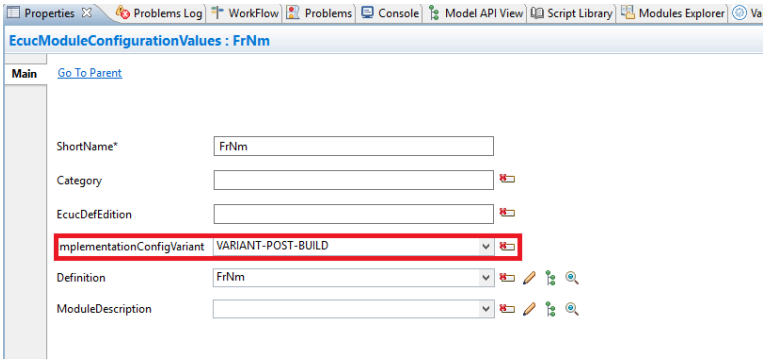


Figure 3.2.Configuration of ImplementationConfigVariant

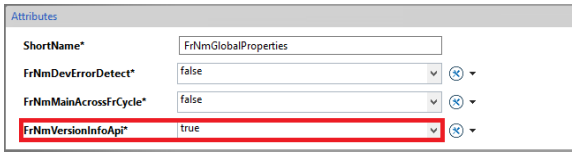
ImplementationConfigVariant will be set to “variant-pre-compile” by default, but if Post-Build Loadable will be used for FrNm, select the “variant-post-build” option.

Step 2 EcuM Configuration

If ImplementationConfigVariant is set to “variant-post-build”, the EcuM configuration for FrNm must be configured with POSTBUILD_PTR.

Figure 3.3.EcuM configuration

And FrNmVersionInfoApi must be set to TRUE.



3.8.5.1 Code Generation Options

Depending on the above setting of ImplementationConfigVariant, the following options can be deployed at Code Generation.

Pre-Compile

If this option is chosen, all ECUs using the generated BSW will have the same configuration and there is no way to change this configuration without re-generating and re-compiling the whole of the ECU's software.

As shown below, select the "FrNm_PreCompile_and_PB_Variant" option for the FrNm module.

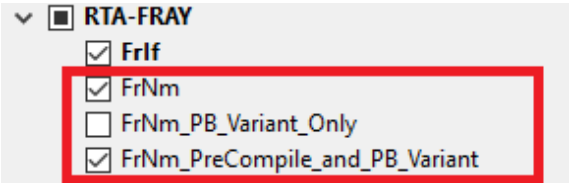


Figure 3.4.Configure full generation all variants

Post-Build Loadable

This option results in the generation of only the configuration for the FrNm module, and it does not generate any of the other code.

Post-Build Loadable is only available after the FrNm module's main code has initialy been fully generated in a Post-Build configurable form, and this is achieved with a one-time Code Generation using the FrNm_PreCompile_and_PB_Variant option, as shown above.

Then, if the module's configuration is subsequently changed, the FrNm_PB_Variant_Only option, also shown above, can be used so that only the configuration files are re-generated (the main FrNm code will not be re-generated).

Note that the "Delete Existing Output Folder" option must be un-ticked. This results in the loadable parts being allocated into a different folder from the full generation that was previously created by the initial full build.

3.9 Integration Advice

3.9.1 Init Functions

FrNm_Init initializes the FrNm module, and the initialization is based on the configuration set pointed to by the nmConfigPtr parameter. If this pointer is NULL_PTR, then the local configuration set available to FrNm will be used for initialization.

Transmission of Vote and Data are stopped indirectly because, after initialization, FrNm will be in Bus Sleep mode.

Reserved bytes will be initialized to 0, user data bytes will be initialized to FF, the Network Request state will be "Released", and all timer variables will be initialized with reset values.

3.9.2 DeInit Functions

There is no dedicated FrNm shutdown functionality.

3.9.3 Scheduled Functions

FrNm_MainFunction_<FrNmChannelRef> is called cyclically to provide maintenance of the FrNm State Machine. As the duration of a FlexRay Cycle may be different for two different Clusters, the calling period is configurable independently for each Cluster at system configuration time.

The main function scheduling is based on the FlexRay cluster ID, which is derived directly from the cluster ID specified in the Nm module network handle information. It is not necessary to specify a cluster ID directly in the FrNm module.

FrNm_MainFunction_<FrNmChannelRef> is the main process of FrNm that is triggered either through a FlexRay synchronous task or an OS task. This will in turn call FrNm_MainFunction that actually implements the FrNm state machine.

If FrNm_Mainfunction is configured as Schedulable, it should be executed at least once in the Flexray cycle, which means that FrNmMainFunctionPeriod should be less than or equal to the FrIf parameter FrIfGdCycle.

3.9.4 General information

- Where possible, use the 'immediate transmission' mode for NM-Vote PDUs, as they might be lost in decoupled transmission mode.
- Ensure that you configure the correct payload length in the FrIf module (using the configuration value `FrIfLSduLength`) when you configure FrNm PDUs. For example, for a pure NM-Vote data PDU, configure the payload

length as 1. For an NM-Data (or combined) PDU, configure the length based on the PDU length configured in FrNm (this equates to the `PduLength` configuration parameter in the global ECU configuration file (EcuC_EcuC-ParamDef.arxml).

- If the FRNM_CYCLE_COUNTER_EMULATION is TRUE, then the network status 'requested' is not considered in NM_STATE_READY_SLEEP state.
- FrNm will pass the complete Tx Pdu to the upper layer(Com) while calling Trigger Transmit. Therefore, Com should ensure that no valid data is sent in the bytes reserved for Standard Bytes.
- If FrNm_Mainfunction() is configured as Schedulable it should be executed at least once in Flexray cycle. Hence FrNmMainFunctionPeriod should be less than or equal to the FrIf parameter FrIfGdCycle. In the same way, if FrNm_MainfuctionTx() is configured as Schedulable, FrNmMainFunctionTx-Period should be less than or equal to FrNmMainFunctionPeriod in order to ensure consistent transmission and reception in respective cycles.

3.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FrNm module uses the following OS resources:

- FrNm_GetPduDataNoNest
- FrNm_GetStateNoNest
- FrNm_GetUserDataNoNest
- FrNm_MainFunctionTxNoNest
- FrNm_RxIndicationNoNest
- FrNm_SetUserDataNoNest
- FrNm_ChangeStateNoNest
- FrNm_RepeatMessageRequestNoNest
- FrNm_RequestSynchronizedPncShutdownNoNest
- FrNm_TxConfirmationNoNest
- FrNm_SetSleepReadyBitNoNest
- FrNm_TriggerTransmitNoNest

3.10 Hardware Requirements

Table 3.26.FrNm Hardware Requirements

Support	Usage
FLASH	3272 bytes
RAM	68 bytes

Usage calculated based on a sample configuration.

4 FlexRay Transport Protocol

Module name	FrTp
Package	RTA-FRAY
AUTOSAR Module ID	36
AUTOSAR Version	22-11

4.1 Introduction

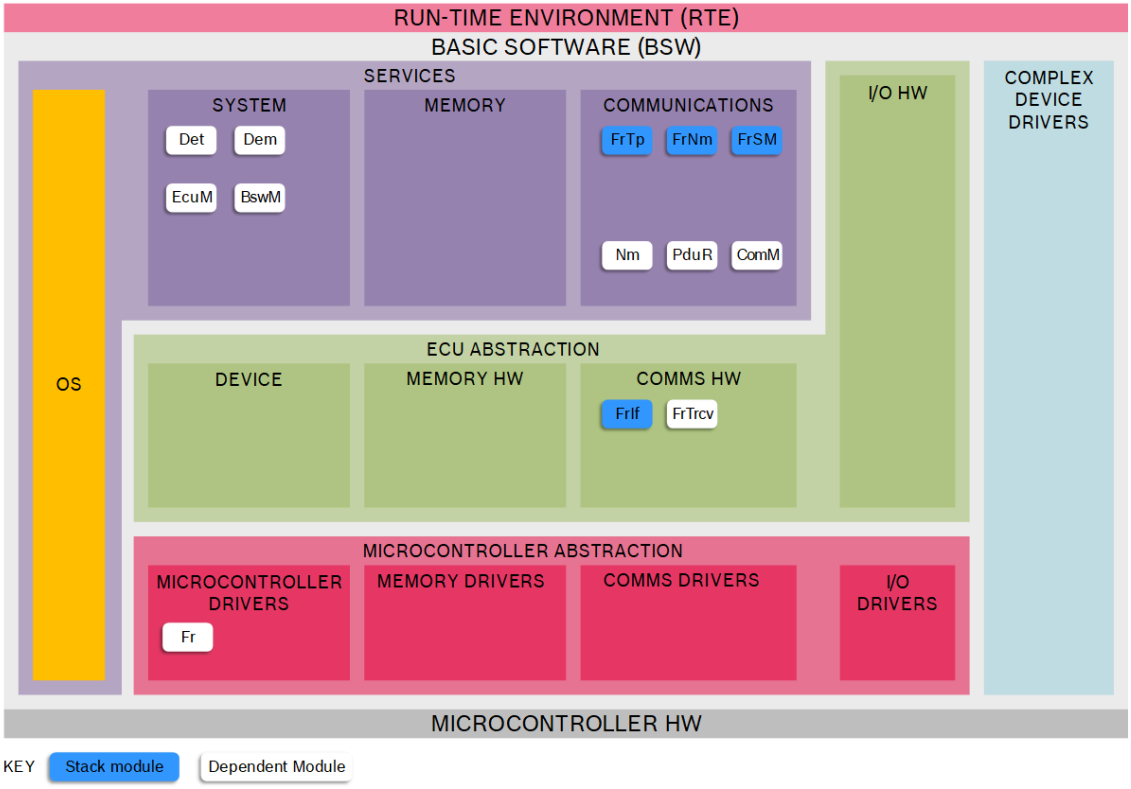


Figure 4.1.FlexRay Architecture

The FlexRay Transport Protocol module (FrTp) is located in the communication hardware abstraction layer, between the communication drivers layer and the upper communication services layer. The diagram above shows the position of FrTp in the AUTOSAR layered architecture.

FrTp receives and transmits FlexRay messages (I-PDUs) between the PDU Router (PduR) and FlexRay Interface (FrIf) modules with flow control. It provides segmentation and reassembly functions for those messages which do not fit within a single FlexRay frame (L-PDU), and it also provides gateway functionality when there is no rate conversion.

The PduR module uses FrTp to route FlexRay diagnostic messages, which typically require segmentation during transmission and reassembly during reception.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

4.2 Supported Features

4.2.1 General

The FrTp module supports the following AUTOSAR-specified functionality:

- Segmentation of data for transmission (with error detection).
- Reassembly of data during reception.
- Data flow control.
- Cancellation of active transmission requests.

The FrTp module fully complies with the ISO-10681-2 standard for message transmission, supporting the following functionality:

- Transmission of a message with a known message length.
- Transmission of a message with an unknown (but finite) message length.
- Acknowledgement of all message transmission operations with a retry mechanism.

4.2.2 Configuration Class

FrTp supports 3 kinds of configuration class -

- Pre-compile
- Post-build Loadable
- Post-build Selectable

See the Configuration Advice for more details.

4.2.3 Additional Functionality

- Schedule Via Joblist Support
- DLL Scheduler

4.2.4 UUDT/RDPI Transmission

FrTp supports UUDT/RDPI transmission, refer to Configuration Advice section for details.

4.3 Extensions

4.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.1.1 FrTp/FrTpGeneral/FrTpRbAPILevelRetrySupport

Short Name	FrTpRbAPILevelRetrySupport
Description	This parameter enabled Retry feature on API level - FrIf_Transmit and PduR_StartOfReception.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.2 FrTp/FrTpGeneral/FrTpRbMaxTxPduCnt

Short Name	FrTpRbMaxTxPduCnt
Description	Maximum possible value of number of uinque FrIf referred Pdus. If this value is to be updated, then source code has to be recompiled.
Multiplicity	0..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.3 FrTp/FrTpGeneral/FrTpRbScheduleViaJoblist

Short Name	FrTpRbScheduleViaJoblist
Description	Preprocessor switch for enabling scheduling of FrTp from FrIf Joblist. Forward JobEntry: Scheduling of FrTp will be done from FrIf Joblist. so FrTp Main Function and FrIfGDCycle should have the same value. when Flexray goes offline scheduling will be done from OS and scheduling from Joblist Resumes once Flexray comes back to Online state.This parameter needs to be enabled only when there is atleast one usecase of transmission on FrTp and all the required parmeters for the transmission path should be configured in FrTp and FrIf as FrTp considers all these parameters for its caluclation. Disable: Scheduling will be done by OS. Enable Callback: only the API FrTp_MainFunction_JobExec will be enabled and joblist won't be created in FrIf by FrTp, this will have to configured manually by the integrator.
Multiplicity	0..1
Default Value	-
Enumeration Literals	FRTP_SCH_FWD_JOB_ENTRY, FRTP_SCH_JOBLIST_DISABLED, ENABLE_CALLBACK
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.4 FrTp/FrTpGeneral/FrTpRbTriggerTransmitSupport

Short Name	FrTpRbTriggerTransmitSupport
Description	This enables Trigger Transmit support on FrTp level. This also uses extra buffers for data storing and transfer. Set this parameter to True even if one of the corresponding FrIf PDU's "FrIfImmediate" parameter is set to False.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.5 FrTp/FrTpMultipleConfig/FrTpConnection/FrTpPduType

Short Name	FrTpPduType
Description	This parameter defines the type of Communication requested from the upper layer. If this parameter is configured as FRTP_IFPDU then FrTp starts the synchronous transmission else normal communication.
Multiplicity	0..1
Default Value	FRTP_TPPDU
Enumeration Literals	FRTP_TPPDU, FRTP_IFPDU
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.6 FrTp/FrTpMultipleConfig/FrTpConnectionControl/FrTpRbRxBfsMax

Short Name	FrTpRbRxBfsMax
Description	This parameter defines the maximum value of BfS that can be sent in the Flow Control during a segmented reception. If this parameter is configured to a non zero value, the Buffer Size Sent in the Flow Control frame from FrTp will be restricted to the configured value, even if complete buffer is provided by the upper layer. This parameter may be used when using a tester which will only accept a fixed value of BfS in the Flow Control frames from the ECU.
Multiplicity	0..1
Maximum Value	65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.7 FrTp/FrTpMultipleConfig/FrTpConnectionControl/FrTpRbTimeBr-

BufferRetryFrTpRbTimeBrBufferRetry

Short Name	FrTpRbTimeBrBufferRetry
Description	This parameter defines the time in seconds that FrTp is allowed to try to acquire a receive buffer. This value forms a part of the total time before which a FC must be transmitted by the receiver (FrTpTimeBr). Therefore, the configuration of this parameter cannot exceed the value of FrTpTimeBr.
Multiplicity	1..1
Value Range	0..0.255
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

4.3.1.8 FrTp/FrTpMultipleConfig/FrTpConnectionControl/FrTpRbTxBfsMax

Short Name	FrTpRbTxBfsMax
Description	This parameter allows the sender to limit the Block size transmitted during a segmented transmission.If this parameter is configured to a non zero value, FrTp will send an CF_EOB frame to the tester after transmitting the configured number of bytes.FrTp will then wait for the next flow control frame to be received from the tester before transmitting the next sequence of frames. This parameter may be configured for use cases where it may be necessary for the ECU to receive a confirmation of the transmitted frames from the tester (by the reception of an FC frame), or, for the use case where the maximum buffer that can be requested from the upper layer, at any point of time , is limited (E.g, in the case of GateWays)
Multiplicity	0..1
Maximum Value	65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.2.1 FrTp_MainFunction_JobExec

Syntax	`FrTp_MainFunction_JobExec(void)`

Description	Used by FrIf to call FrTp Main Function from its Joblist callback function.

4.4 Limitations

- Retry is not supported if TxConfirmation is not received.
- AUTOSAR requirements [SWS_FrTp_01077], [SWS_FrTp_00418], [SWS_FrTp_00421] and [SWS_FrTp_01074] are not supported.
- When FrIf_Transmit returns E_NOT_OK, the retry mechanism is done through FrTpTimeCs (instead of FrTpTimeFrIf and FrTpMaxFrIf).
- In the Post-Build Loadable variant, note that only parameter values can be updated (i.e. No addition of containers is possible at Post-Build time).
- 16-bit Micro controller architecture is not supported as the Compiler abstraction is not implemented.

4.5 Exclusions

There are no unsupported API function calls in this release.

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- FrTp/FrTpGeneral/FrTpUnknownMsgLength
- FrTp/FrTpMultipleConfig/FrTpConnection/FrTpBandwidthLimitation

4.6 Deviations

4.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

4.7 API Definition

List of supported API functions

Table 4.11.FrTp Supported API functions

`FrTp_Transmit`	Requests transfer of an SDU and sets a flag indicating the request.
`FrTp_Init`	Initialises (or re-initialises) the module.
`FrTp_Shutdown`	Closes all pending connections and stops the module.
`FrTp_ChangeParameter`	Changes a specified parameter's value for an identified I-PDU.
`FrTp_CancelReceive`	Cancels the reception of an SDU.

`FrTp_CancelTransmit`	Cancels the transmission of an SDU.
`FrTp_GetVersionInfo`	Returns the vendor ID, module ID and version info for the module.
`FrTp_RxIndication`	Callback used by lower layer communication interface module to indicate a received I-PDU.
`FrTp_TxConfirmation`	Callback used by lower layer communication interface module to confirm the transmission of an IPDU.
`FrTp_TriggerTransmit`	Callback used by the upper layer module to check whether the available data fits into the buffer size reported by PduInfoPtr→SduLength.
`FrTp_MainFunction`	The central processing function of FrTp.

For Further information please refer to AUTOSAR_SWS_FlexRayISOTransportLayer.pdf

4.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

4.8.1 Code Generation based on configuration class

FrTp supports 3 kinds of configuration class -

- Pre-compile
- Post-build Loadable
- Post-build Selectable

If either of the two Post-Build options is to be used for FrTp, the following two configuration steps must first be completed.

Step 1 Select the Configuration Class

Go to the FrTp module configuration, and then the "Properties" tab.

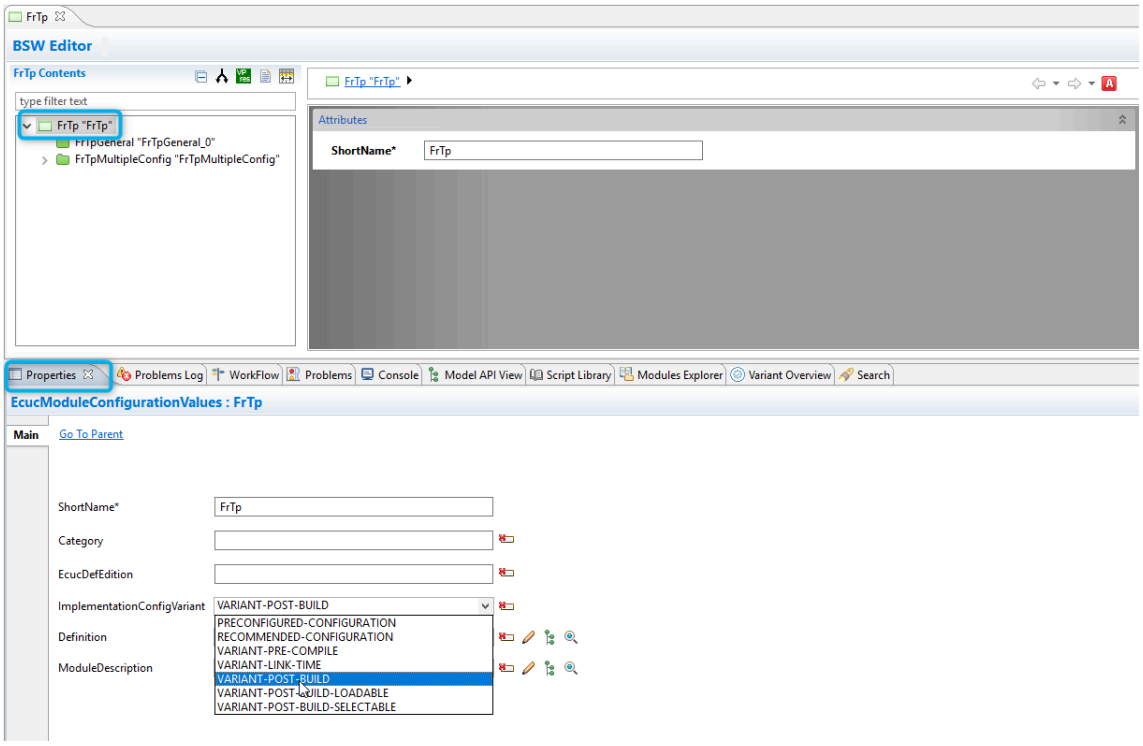


Figure 4.2.Select which configuration class of FrTp

ImplementationConfigVariant will be set to “variant-pre-compile” by default, but if either of the two Post-Build options will be used for FrTp, you must select the generic “variant-post-build” option (and not the specific “variant-post-build-loadable” or “variant-post-build-selectable” options).

Step 2 EcuM Configuration

If ImplementationConfigVariant is set to “variant-post-build”, the EcuM configuration for FrTp and all its dependent modules must be configured with POST-BUILD_PTR.

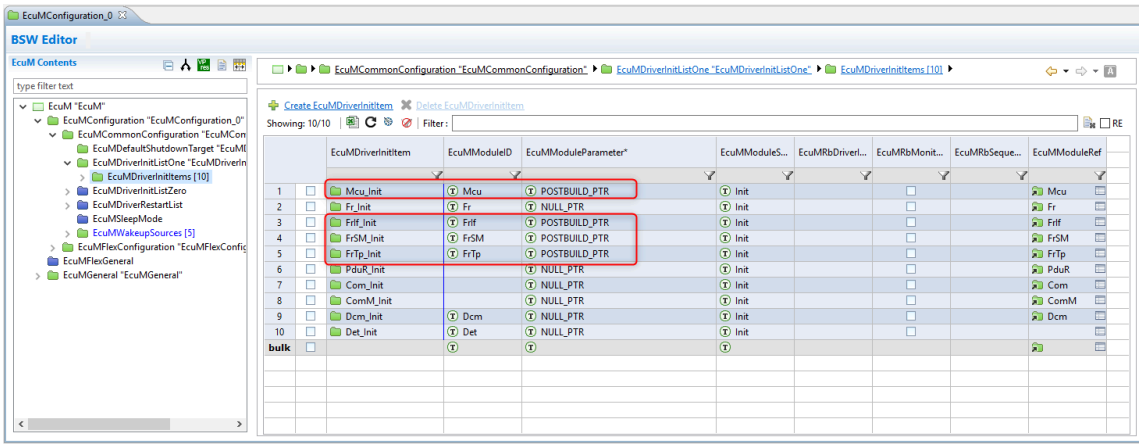


Figure 4.3.EcuM configuration

4.8.1.1 Code Generation Options

Depending on the above setting of ImplementationConfigVariant, the following options can be deployed at Code Generation.

Pre-Compile

If this option is chosen, all ECUs using the generated BSW will have the same configuration and there is no way to change this configuration without re-generating and re-compiling the whole of the ECU's software.

As shown below, select the "FrTp_PreCompile" option for the FrTp module. The other options on the right can be set at your discretion.

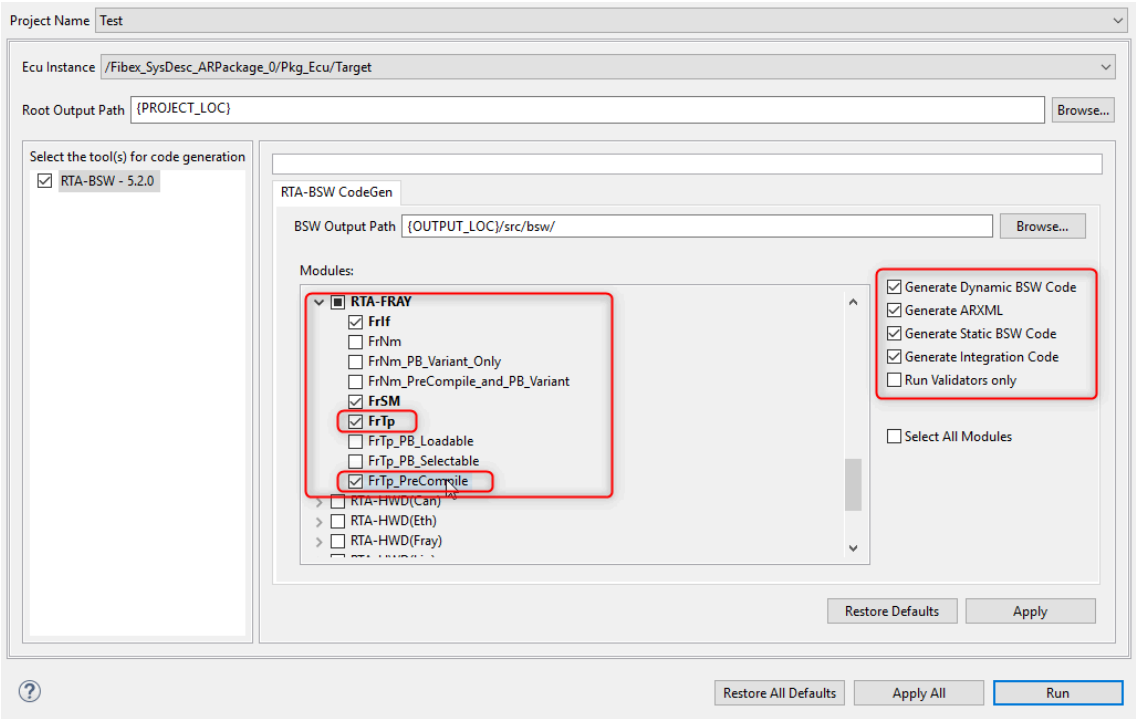


Figure 4.4.Generate FrTp with Configuration Class as Pre-compile

Post-Build Loadable

This option results in the generation of only the configuration for the FrTp module, and it does not generate any of the other code.

Post-Build Loadable is only available after the FrTp module's main code has initially been fully generated in a Post-Build configurable form, and this is achieved with a one-time Code Generation using Post-Build Selectable, as shown here.

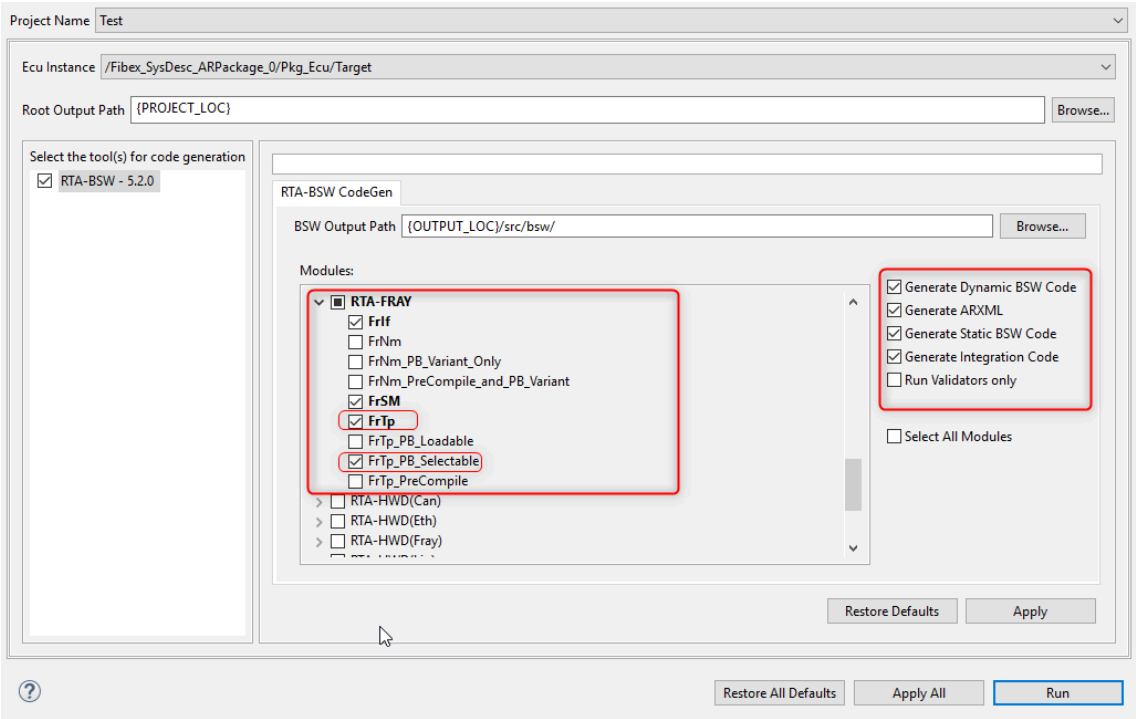


Figure 4.5.Full Code Generation with Post-Build

Then, if the module's configuration is subsequently changed, as shown below the Post-Build Loadable option can be used so that only the configuration files are re-generated (the main FrTp code will not be re-generated).

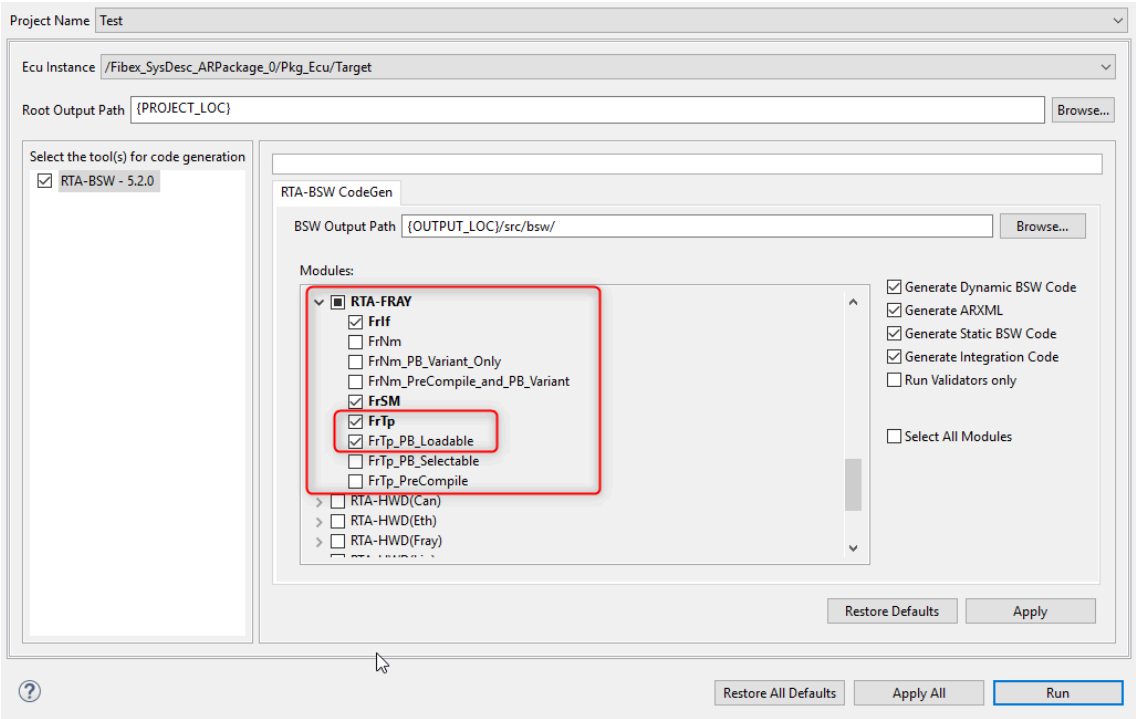


Figure 4.6.Regeneration for Post-Build

Note that the “Delete Existing Output Folder” option must be un-ticked when using Post-Build Loadable. This results in the loadable parts being allocated into a different folder from the full generation that was previously created by the initial Post-Build Selectable.

Post-Build Selectable

This option is used when multiple configurations for FrTp will be created, and the required configuration can be selected at run time.

As shown below, the FrTp_PB_Selectable option is ticked.

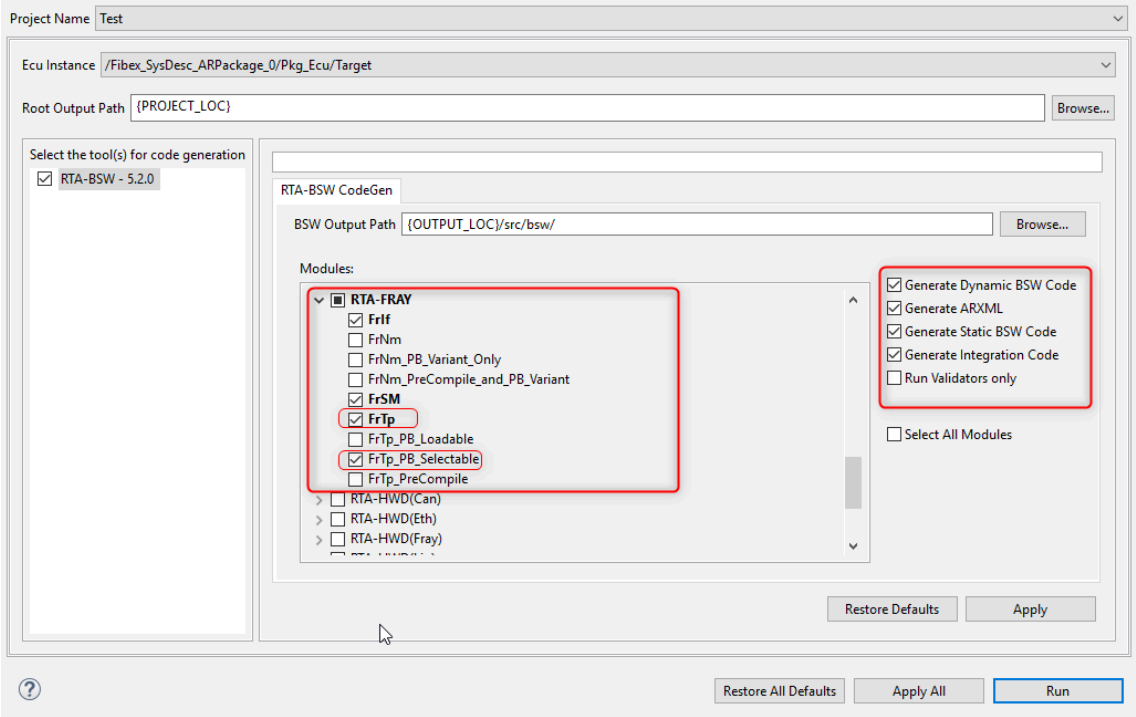


Figure 4.7.Code Generation with Selectable Post-Build

4.8.2 UUDT/RDPI Transmission

FrTp supports UUDT/RDPI transmission:

- Configuring the FrTpPduType parameter as FRTP_IFPDU.
- For UUDT transmission, the parameter FrTpMultipleReceiverCon shall be set to TRUE. The UUDT transmission doesn't support the Retry Ack feature.

4.9 Integration Advice

4.9.1 Init Functions

FrTp_Init has to be called during the startup phase of the ECU, and it is also used to reinitialize FrTp after an FrTp_Shutdown has been called. Before Initialization, at OFF state, FrTp only allows an update of Postbuild parameter through FrTp_Init.

4.9.2 DeInit Functions

FrTp_Shutdown shuts down the FrTp module.

4.9.3 Scheduled Functions

FrTp_Mainfunction should be called cyclically, or by FrIf using the Schedule Via Joblist option.

4.9.4 Integration with Dcm

Note the following points when integrating FrTp with the Dcm module:

- FrTp main functions should be called with a period faster than Dcm (`FrTp/FrTpGeneral/FrTpMainFuncCycle` should be a shorter period than `Dcm/DcmGeneral/DcmTaskTime`). This eliminates the risk of losing data when there are too many requests from Dcm.
- Dcm_MainFunction must be called prior to FrTp_MainFunction (see `PositionInTask` parameter in `RteEventToTaskMapping`). This ensures that the FrTp channel state is changed to idle (by FrTp_MainFunction) before Dcm_-Mainfunction triggers sending data.

4.9.5 Integration with EcuM

While configuring FrTp_Init in the `EcuMDriverInitListOne` container of EcuM, choose the value "NULL_PTR" or a valid CfgPtr for `EcuMModuleParameter`.

4.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The FrTp module uses the following OS resources:

- FrTp_EXCLUSIVE_AREA

4.10 Hardware Requirements

Table 4.12.FrTp Hardware Requirements

Support	Usage
FLASH	8128 bytes
RAM	97 bytes

Usage calculated based on a sample configuration.

5 Glossary

CC FlexRay Communication Controller

Cluster
A communication system made up of a number of nodes linked by a physical transmission medium

FrIf AUTOSAR FlexRay Interface module

FrNm
AUTOSAR FlexRay Network Management module

FrSM
AUTOSAR FlexRay State Manager module

FrTp
AUTOSAR FlexRay Transport Layer module

L-PDU
PDU of the data link layer

Node
An ECU which is connected to a FlexRay bus via the FlexRay Interface

PDU
Protocol Data Unit